Grammaticality de-idealized

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Abstract

This paper introduces the Operator-Value Model of Grammaticality (OVMG), a framework for (un)grammaticality distinct from grammars focused solely on language production or description. The model treats (un)grammaticality as conditioned stability in form–value relations accepted or rejected within communicative situations, integrating mapping viability, operator-value coherence, population licensing, and speaker-level feelings of anomaly. Previous approaches, including generative grammar, Construction Grammar, and psycholinguistic models, have each captured part of the picture but haven't jointly explained why semantically coherent constructions may nonetheless be deemed ungrammatical, why certain transparent structures remain systematically unentrenched, and why categorical grammaticality clusters where it does. The formal core treats grammatical status as a posterior over licensed assemblies: operator-value compatibility is hard (a unification condition, not a weighted penalty), obligatoriness is derived from preemption of zero-marking rather than stipulated per language, and categoricality emerges as bimodality of the population licensing state under preemption and repair dynamics, rather than from structural membership or intuition alone. The subjective phenomenology is a two-channel read-out–anomaly and confidence–which derives the contrast between the confident rejection of preempted gaps and the ineffability of defective paradigm cells. It earns its keep by projection: it predicts which rejected constructions should soften under repeated exposure and which should resist, how learners should treat zero evidence under different opportunity structures, how early second-language rejections should track first-language preemption, where community treatment of a form should change discontinuously as licensing crosses a situational threshold, and where judgment dispersion should rise before means move as a contrast goes moribund.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

Introduction

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

(1) a. **Can the have running?*
b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
c. **I've finished it yesterday.*
d. *?I saw Joan, a friend of whose was visiting.*
e. *The bread the baker the apprentice helped made is delicious.*
f. A: *How old are you?* B: **I have 25 years.*
g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve a clash between the time semantics of clause tense and the word *yesterday*. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called theoretically predicted acceptability: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are transparent and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different traditions emphasize different pieces: formal approaches focus on abstract principles, usage-based theories on frequency and entrenchment, processing accounts on cognitive constraints. Each captures part of the picture, but none unifies categorical constraints, gradient acceptability, cross-linguistic variation, and change over time. I'll argue that the obstacle is a shared idealization, and that removing it reorders the questions.

The de-idealization in the title is specific. Early generative grammar idealized grammaticality to the knowledge of an ideal speaker–listener in a completely homogeneous speech community (Chomsky, 1965, p. 3), treating variation, uncertainty, and change as noise around a categorical competence fact. That idealization bought tractability, but it priced out the phenomena in (1). Following Goodman (1955), removing it reorders the questions: ask first what the category GRAMMATICAL lets us predict (its projectibility, what observing some features licenses us to expect about others), and only then what worldly profile supports those predictions and what stabilizes it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Mechanisms matter because they explain why the projection holds, not because naming them settles the kind.

This paper develops that answer as the Operator-Value Model of Grammaticality (OVMG). Its unit is the OPERATOR value: a closed-paradigm, update-configuring contrast (tense, agreement, clause type, and their kin), identified by its public-update role rather than by expressive substance, so that tone, particles, word order, and inflection all count when they realize such a contrast (Reynolds, 2026c). The framework rests on three premises, which track the order just named: profile, projection, and support.

1. Profile. Grammaticality is a conditioned form–value relation, specified by whether an operator value is structurally realizable, coherent, and licensed within a communicative situation (community, dialect, register). This profile is disturbed in distinct ways (no licensable analysis, clashing construals, absent situational licensing, processing overload), so the phenomena traditionally lumped as "ungrammatical" in (1) fall into separable modes rather than one kind.
2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.
3. Support, graded. The categorical appearance of grammaticality emerges when usage stabilizes the population's licensing state; conversational repair supplies a candidate *controller* for that state, testable by its intervention signature (§4.3); and the subjective feeling of ungrammaticality is a separate, gradient signal–an anomaly–confidence pair–not grammatical status itself.

Together, the premises explain why the traditions have talked past one another. The profile premise preserves the formal insight that constraints can be systematic while letting categorical and gradient cases share one architecture, in line with Francis's (2022) case for building gradience into core grammatical theory. The projection premise keeps the account testable: low-evidence constructions should respond to exposure and framing differently from high-opportunity preempted forms. The support premise explains why formal licensing, processing cost, and subjective feeling can move together without being the same thing.

Section 1 traces the impasse in grammaticality theory to the attempt to treat it as one phenomenon. Section 2 presents OVMG, defining the constitutive conditions and the recurrent instability modes that route a form to one diagnostic category or another. Section 3 gives the formal core: a state theory that specifies grammatical status as a posterior over licensed assemblies, and a dynamics module (§4) that shows how such states arise, stabilize, and–where the repair evidence warrants–are actively maintained. Together they derive categoricality and obligatoriness from usage rather than stipulating them. Section 5 situates the account against generative grammar, Construction Grammar, usage-based, logicality, and relevance-theoretic approaches, and states the predictions. The paper closes with limitations and a research program spanning corpus, experimental, and cross-linguistic methods.

1 The impasse in grammaticality theory

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a unified account of what makes an utterance grammatical or ungrammatical. The impasse persists because grammaticality is often treated as a single phenomenon, when it actually arises through the interaction of three distinct sources of failure. Understanding why previous approaches have struggled requires disentangling three components: structural viability, interpretive coherence, and situational licensing.

1.1 The problem of structural viability

The first challenge in defining grammaticality is distinguishing structures that admit no viable analysis from those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. That categorical view captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The "categorical" view correctly identifies that a viable structural analysis is necessary for grammaticality. But by raising this single condition to the *definition* of grammaticality, formal approaches struggled with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts. The OVMG framework resolves this by recognizing that while structural viability (the successful mapping of form to category) is a necessary, categorical prerequisite, it isn't the sole determinant of grammaticality.

1.2 The problem of interpretive coherence

The second tension arises from the interaction of form and meaning. Chomsky's famous example (1b) shows that operator-valued form–value relations can be intact even when compositional semantics fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the compositional semantic relation: the lexical values don't cohere. Conversely, (1c) **I've finished it yesterday* shows compositional semantics can be transparent (the intended meaning is clear) while an operator-valued form–value relation fails. Here, the clash is between the present perfect's temporal value and the adverb's deictic anchoring.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash (like **She disappeared him the evidence*) are rejected. A complete theory needs to account for interpretive coherence: the degree to which a structurally viable form yields a stable, non-contradictory interpretation.

1.3 The problem of situational licensing

Finally, even structurally viable and semantically coherent forms may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011): speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

(3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and semantically transparent (the intended meaning is clear), but it's categorically rejected in English. The same form is grammatical in French (*J'ai 25 ans*) and Spanish (*Tengo 25 años*); the constraint is community-specific. Usage-based approaches (Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, leaving this form without positive evidence.

These three problems (structural viability, interpretive coherence, and situational licensing) correspond exactly to the three constitutive variables of the OVMG framework presented in Section 2. Acknowledging them as distinct but interacting components makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single "competence" definition or a purely probabilistic usage model.

2 The Operator-Value Model of Grammaticality

The term VALUE is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. This isn't feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The model's name marks its unit: operator values (closed-paradigm, update-configuring contrasts). A companion operator-stratum paper develops this boundary in full and argues that it follows public-update role rather than expressive substance: tone, particles, word order, inflection, and suprasegmental material all fall within its scope when they realize an operator contrast (Reynolds, 2026c).

Here a self-contained working test suffices, because the concept has to stand on its own inside this paper: it fixes the scope of hard compatibility, the categoricality prediction, and the appendix on Turkish suffix harmony (§A).

A contrast is an operator contrast when it meets three conditions jointly: drawn from a *closed* paradigm, *high-opportunity* (the communicative job recurs constantly), and *configured to the public update* (clause type, polarity, tense and aspect anchoring, agreement and role allocation, scope, or reference tracking). Mis-setting such a contrast changes the update instruction even when the intended message is recoverable. The three conditions have formal counterparts in §§3–4: closure is finiteness of the dimension's value set, opportunity is the niche count $N_t(n, c)$, and update-configuration is a positive expected common-ground divergence $\Delta(d) > 0$ from mis-setting the contrast (Reynolds, 2026c).

These conditions exclude cases that "closed-class", "grammaticalized", or "high-frequency" alone would admit. A closed-class item that doesn't configure the update, such as a purely stance-marking discourse particle, fails the third condition and is predicted *not* to attract categorical policing; a high-frequency open-class collocation fails the first; and a genuinely update-configuring contrast that's gone low-opportunity, such as a moribund case distinction, fails the second and is predicted to destabilize. The prediction is falsifiable in the direction that matters: a closed, high-opportunity contrast that isn't update-configuring should be policed like style, not like grammar.

The formal architecture rests on four constitutive conditions: structural viability, operator-value compatibility, saturation of obligatory dimensions, and situational licensing. Together they specify a

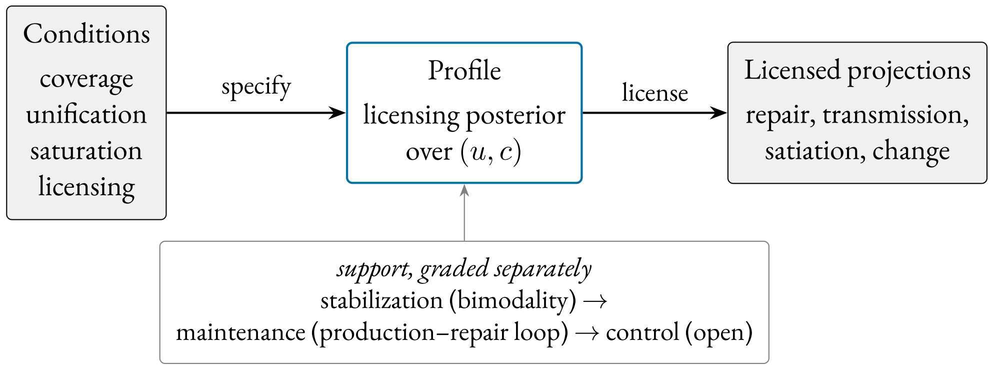

Figure 1: The projectibility-first order of explanation. The constitutive conditions specify a profile (the posterior over licensed assemblies) whose partial observation licenses the projections on the right. What stabilizes, maintains, or controls the profile is a separate, graded claim (below), not something inherited from naming the mechanism.

PROFILE in the projectibility-first sense: a pattern whose partial observation licenses further expectations (Reynolds, 2026a). Their interaction determines grammaticality (§3); what the profile projects, and what supports those projections, is the business of §3 and §4. Figure 1 lays out this order of explanation, which the rest of the paper fills in. Table 1 summarizes these conditions; the formal definitions appear in §3.

Condition	Component
$\mathcal{A}(u, v) \neq \emptyset$	Structural viability (categorical prerequisite)
$\mathrm{def}(A)$	Operator-value compatibility (unification)
$\mathrm{sat}(A, c)$	Saturation of obligatory dimensions (derived, §4.5)
$L_t(A, c)$	Situational licensing (latent, population-level)

Table 1: The constitutive conditions of the OVMG framework. Their conjunction, read through the learner's posterior, gives grammatical status (§3.3.1).

The formal core needs names for recurrent INSTABILITY MODES: ways of defeating coverage, unification, saturation, or licensing, with processing and prescriptive factors modulating how grammatical something *feels* without directly fixing grammatical status.

2.1 Diagnostic categories

The categories below are labels for common routes to ungrammaticality, not primitives of the model. Each traces to one or more constitutive variables, as the formal core in §3 makes explicit.

FORM–VALUE RELATIONS. Grammaticality depends on stable links between forms and the operator values they realize within communicative situations. That form–value relations exist at every level of grammatical organization, including morphemes and abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). OVMG narrows the target of categorical grammaticality to closed-paradigm, high-opportunity, update-critical contrasts: resources that configure how an utterance enters public coordination.

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgarriff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow…*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence for this integration of form and meaning at the processing level. Fedorenko and colleagues find that brain regions activated by syntactic manipulations overlap substantially with those activated by semantic ones (Fedorenko et al., 2011, 2012, 2024). The overlap points to partly shared processing resources rather than wholly separate modules.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a "communicative situation". The emergence claim defended below is narrower: categoricality arises through the licensing dynamics in §4.4, not through an unanalyzed appeal to usage.

Communicative situations aren't static or mutually exclusive. A speaker orients to overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they're orienting to (which can shift with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions.

Early child language shows this. Toddlers produce utterances that deviate from adult norms but remain internally consistent within the child's developing system. A toddler in a monolingual English household might say:

(4) *Ava cookie.* (intended as 'Ava = I want a cookie')

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child's regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

(5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar pattern appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community's norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

(6)

Ayer,	*estábamos*	*lifting*	*en*	*el*	*gym*	*durante*	*una*	*hora.*
yesterday	be.IMPF-1PL	lifting	in	the	gym	for	an	hour

'Yesterday, we were lifting in the gym for an hour.'

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, in a standard monolingual Spanish setting, where fully Spanish progressive structures are licensed (*estábamos levantando pesas*), speakers may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn't mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

(7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
the dwarfs try.PST-3PL but not succeeded in awakening Snow White
'The dwarfs tried but didn't succeed in awakening Snow White.'

In a slightly different case, as a second-language speaker of Japanese, I used to say

(8) * *Kawaii da*.
cute-PRES COP.PRES
'(That)'s cute.' (intended)

Conventionally, though, the redundant tense marking has no accepted meaning, making the use of the copula ungrammatical to most Japanese speakers, though it felt meaningful and grammatical to me. The example shows why acquisition history matters: shared routines stabilize form–value relations for a community, while individual trajectories can still pull judgments away from that norm.

The specific linguistic context can matter too:

> To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)[1]

These examples show why grammaticality has to be indexed to communicative situation. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical, and both may be right within their respective contexts. The linguist's task is to determine whether a form is grammatical from some position in that space, or systematically excluded from all of it.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions.

[1] The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

In different languages, different distinctions become conventionalized as grammatically obligatory. These obligatory contrasts reflect historically stabilized patterns in which meaning distinctions are systematically marked. Over time, usage establishes grammatical constructions that reliably signal these distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it isn't optional but an obligatory grammatical marker for ongoing, incomplete actions:

(9) *She is studying right now.*

Here, the progressive form *is studying* isn't just a stylistic choice. It's the recognized, grammatical way to convey the appropriate aspectual meaning in this frame.

In French, by contrast, the progressive aspect isn't a grammatically mandatory distinction. Speakers can signal ongoing activity through adverbs or periphrastic constructions, but standard French doesn't have a dedicated progressive form. The sentence:

(10) *Elle étudie maintenant.*
'she studies/is studying now'

can comfortably describe a currently ongoing action without any need for special morphology. In Standard French, this aspectual distinction isn't conventionalized as an obligatory operator contrast. English grammar treats the contrast as obligatory; French doesn't. On this account that difference isn't a primitive of either grammar: §4.5 derives obligatoriness as the limit of preemption applied to zero-marking, so the English–French contrast is a prediction of the dynamics, not a stipulation they inherit.

The same holds for evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically realized through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

(11) *Gelmiş*
'He/she came (apparently)'
(i.e. the speaker wasn't a witness but inferred or heard about it.)

In Turkish, evidential distinctions are conventionalized as obligatory operator contrasts. A speaker can't simply omit evidential marking without sounding ungrammatical.

English speakers can say *I heard that he arrived* or *He must have arrived*, but these are optional lexical or modal resources rather than required elements of the grammar. In English, evidential distinctions aren't conventionalized as obligatory operator contrasts. The same derivation covers the Turkish case on a different dimension (§4.5).

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can grammaticalize quite different semantic distinctions as obligatory. The contrast reflects different conventions about which meaning distinctions are systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is necessary for grammaticality, but it isn't sufficient. Speakers routinely interpret strings that still count as ill-formed in the relevant communicative situation. That flexibility makes licensing, not interpretability in principle, the central question.

Three further pressures sort candidate mappings: conflict with conventional constructional values contributes negative evidence; high recovery cost down-weights the mapping; and weak conventionalization leaves the mapping with insufficient positive evidence. Their interaction predicts why **Furiously sleep ideas green colorless* elicits universal rejection while (1e) feels bad in everyday quotation but is accepted as "grammatical but hard to process" once speakers are walked through the intended structure. The weighting concentrates probability mass on the smaller subset of imaginable relations recognized as bona fide grammatical resources.

Nor does convergent usage by itself settle the matter. Defining grammaticality as "whatever speakers accept" would be circular: it couldn't flag contested innovations, explain stable dialectal differences despite mutual intelligibility, or distinguish licensed forms from performance errors that pass unnoticed. Separating mapping, compatibility, and licensing avoids trivializing grammaticality while still treating durable population-level licensing as decisive for which relationships endure.

Operator-value compatibility. Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations realize. This component captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I've finished it yesterday.*

Here, the present perfect construction realizes a current-relevance operator value while *yesterday* specifies completed past time. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows a direct conflict between the temporal value realized by the grammatical construction and the lexical temporal specification.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as 'I'm 25 years old'

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is transparent, so the failure is not an operator-compatibility failure. It is a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes value incompatibility involves conflicting information-structural requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

The clash is in information structure: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit compatibility. The formal core treats operator-value conflict as failed unification among partial operator assignments: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-structural requirements where a construction grammatically packages focus or backgrounding, and indexical contrasts only when they have entered a closed accountable repertoire. This makes compatibility partly derivative of licensing facts at the feature level, keeping arbitrary lexical implausibility out of grammatical incoherence.

Processing constraints. Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems, particularly through multiple long-distance dependencies or heavy embedding, they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

Dependency locality is the best-documented such cost. Integration difficulty rises with dependency length and peaks at loci of long-distance integration, with locality accounting for 59% of region-level variance in reading times in one experiment and 73% in another (Grodner & Gibson, 2005); the figure is considerably lower once item-specific variance is included. Because the effect is so robust, the formal treatment in §3 gives dependency locality a dedicated component rather than hiding it inside a residual term. High-locality constructions feel worse even when structurally well-formed, and because they're harder to produce and parse, speakers tend to avoid them.

That avoidance is a pathway from processing to change, developed formally in §2.5 and §4: lower-processing forms are easier to store and reuse, so simpler patterns can spread and conventionalize. The processing evidence supports this as a plausibility claim about one route into stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with other components. A construction that's marginal due to low situational acceptance may become completely unacceptable when combined with processing difficulty. Conversely, highly entrenched constructions may remain acceptable despite considerable processing demands; strong conventionalization can partially offset processing costs.

Socio-pragmatic indexicality. The value of a construction extends beyond semantic content to include socio-pragmatic dimensions: how linguistic forms index aspects of social context, speaker identity, group membership, stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

(16) *¿Vos querés un café?*
you.SG want.2SG-VOS a coffee
'Do you want a coffee?'

Here, the use of *vos* rather than *tú* denotes the second-person singular hearer (the person being offered a coffee) and indexes the speaker's regional identity and familiarity with the local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors; the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn't depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

Phonology can carry constructional indexical value, but it bears on grammaticality only when an operator value conflicts. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn't trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, and some participants "hallucinated" identity-linked features that weren't present in the signal.

So a construction's value reaches beyond semantic features to socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions helps explain why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

Situational entrenchment and acceptance. Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized acceptance. This component captures the degree to which a form–value relation has become entrenched within relevant communicative situations.

(17) * *We sheared three sheeps.*

The regular plural marking is semantically transparent and structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational entrenchment operates independently from the other components. A construction may have perfect value compatibility and minimal processing demands but still be rejected because a different form has been conventionalized for that meaning. Extreme rarity shows the point clearly. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions (sufficient accessibility of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis), which are seldom met all at once (Reynolds, 2024):

(18) ? *I saw Joan, a friend of whose was visiting.*
(adapted from Huddleston & Pullum 2002, p. 472)

In the COCA check reported here, no token of the construction appears (Davies, 2024). That absence supplies little negative evidence, because the opportunity set is tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers have little evidence either way: neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The evidence therefore supports high *uncertainty* rather than confident rejection. Some speakers can draw an analogy with similar constructions and accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s account predicts that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts sharply with preempted cases like left-branch extraction, discussed next, where the opportunity set is large and the construction is systematically untaken, giving speakers overwhelming evidence that it isn't licensed.

Apparent categorical exclusions. Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy* [__ *car*]*?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of "ungrammaticality".

Under OVMG, such patterns are analyzed not as cases that require a separate constraint on possible structure but as preempted gaps: situational licensing is driven toward zero by persistent preemption. Because the communicative job is common and an established competitor always wins (e.g., *Which car did you buy?*), speakers accumulate overwhelming evidence that the construction isn't part of the repertoire. Additional processing penalties, such as systematic garden-path reanalysis when the bare interrogative phrase is encountered, can strengthen the subjective categoricality without any independent structural constraint being posited. The formal dynamics appear in §3.

Cross-linguistic comparison doesn't threaten this analysis, because the dynamics are defined over a language's own niches. Left branch is a descriptive class tied to a particular phrase-structure analysis, not a comparative concept (Haspelmath, 2010); whether discontinuous nominals in case-rich, article-less languages instantiate the same configuration as the English case is far from clear, and the companion corpus study deliberately brackets the comparison (Reynolds, 2026d). What the model inherits instead is a prediction about opportunity structure: superficially similar discontinuous-nominal patterns should be learnable where inflectional marking makes the dependency recoverable, a productive source of discontinuous order supplies scaffolding, and a discourse niche makes the separated element useful. English meets none of the three, so the candidate has no scaffolding of its own and the contiguity-preserving competitors win every opportunity.

Several diagnostic criteria help identify these preempted-gap constructions:

1. Persistent unacceptability: The construction never gains acceptance; familiarity and repetition don't shift judgments.

2. Categorical rejection: Speakers find the constructions simply impossible, with no intermediate ratings.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If participants are led to treat presented tokens (like LBE) as intentional dialectal or ingroup resources rather than errors, then satiation (or at least reduced feeling of ungrammaticality) should be more likely than in standard paradigms where the construction is treated as simply wrong.

If a construction resists even this preempted-gap analysis (a form for which no structural analysis is available at all), it would mark the residual case for an independent constraint on possible analysis. Treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don't. They leave a high-opportunity cell empty without any single form winning it: most English varieties lack a negated contracted form for *am* (**amn't*, though Irish English and some Scottish varieties license it), and Russian speakers avoid the expected first-person singular non-past of *pobedit'* 'win', resorting to periphrasis instead (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

The niche is common and candidate forms are transparent, but no variant accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving each candidate's posterior low in mean but unstable, rather than concentrated near zero.

The phenomenology matches: speakers report not knowing how to say it (ineffability), not the confident "that's wrong" of a preempted gap. §4.6 derives this profile from the production model itself: avoidance absorbs the cell's opportunities, and because avoidance is nearly uninformative about any particular candidate, no candidate accumulates confident negative evidence either. The satiation-framing prediction follows: defective cells should behave like low-evidence forms, movable under framing, not like preempted gaps.

2.2 DIAGNOSING (UN)GRAMMATICALITY AT A GLANCE

For exposition the recurrent instability modes can be read as a short decision tree (the formal definitions appear in §3):

Condition	⟶ Canonical outcome
no viable structural analysis	⟶ NONSENSE (**Can the have running*)
operator-value clash (unification failure)	⟶ value incompatibility (**I've finished it yesterday*)
rarely encountered, low certainty	⟶ community-novel (*friend of whose*)
systematically preempted	⟶ preempted gap (left-branch extraction)
high opportunity, candidates split	⟶ defective cell (**amn't*)
high processing cost, structure recoverable	⟶ transient ill-formedness (centre embedding)
otherwise	⟶ grammatical

Lexical-semantic oddity without an operator clash (*colorless green ideas*) routes to "grammatical"; its strangeness is a construal-plausibility cost in the feeling channel (§3.7), not a status fact.

2.3 Patterns of (un)grammaticality

When these components interact, constructions succeed or fail in systematic but non-binary ways. The diagnostic categories above name the main routes to failure; the decision tree summarizes them. Forms also differ in how far they fall from licensing.

Degrees of violation. The degree of violation reflects the scope and intensity of component mismatches. Violations can range from mild, easily recoverable deviations to complete breakdowns in interpretability.

Minor violations might involve a single component with partial conflict, as when a construction has moderate processing difficulty but full value compatibility and strong situational entrenchment. Such cases often receive intermediate acceptability ratings and may improve with exposure.

Severe violations typically involve multiple components or complete failure of a single critical component. In a construction lacking any viable form–value relation, the breakdown is total, while violations of categorical constraints produce consistent, strong rejection regardless of other factors.

The interaction between components can amplify or mitigate violations. Strong situational entrenchment can partially offset processing difficulty, while semantic transparency can make marginal syntactic patterns more acceptable. Conversely, multiple minor violations can compound to produce strong ungrammaticality judgments.

2.4 The subjective experience of grammaticality

The constitutive variables determine conventional grammatical status (§3.3); speakers register putative violations through feelings and judgments that can diverge from it. [2]

The feeling of ungrammaticality. The subjective feeling of ungrammaticality is a metacognitive response to linguistic violations. It parallels other well-studied metacognitive feelings like the feeling of knowing (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

There isn't a positive feeling of grammaticality, just as there isn't a feeling of having enough oxygen, only the negative feeling registered in its absence. The feeling of ungrammaticality, then, can be seen as the negative response triggered when an utterance violates expected form–value patterns.

This notion echoes Edward Sapir's "form-feeling", an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the same separation. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

Unstable or missing form–value relations, detected during processing, can trigger this response. The response is a speaker-level heuristic for detection, not the definition of grammaticality. That separation explains why marginal constructions elicit gradient certainty, why repeated exposure can

[2] The distinction between grammatical status and the subjective feeling has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it, bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. For LLM-based grammaticality research, the distinction blocks an easy inference: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying or merely process patterns that correlate with those norms remains an open question (Techio, 2026).

attenuate negative responses without necessarily changing status, and why objectively licensed constructions can still feel wrong under processing pressure. In that sense, (un)grammaticality can be illusory (Fanselow, 2021).

Distinguishing grammatical status from subjective ratings. Grammaticality research has to distinguish grammatical status, the conventional, population-level property made precise in §3.3, from the subjective feelings and ratings that speakers provide. (I use "objective" below only in that sense, population-level and convention-constituted, not judge-independent.)

Table 2 summarizes the two levels of analysis.

Table 2: Grammatical status versus subjective experience

Level	Nature	Observable through
Grammatical status	Whether a form–value relation is licensed in the communicative situation	Converging evidence: corpora, production, repair behaviour
Subjective feeling	Metacognitive warning signal that "something is wrong here"	Directly observable: ratings, eye-tracking, self-reports

The measurement problem is familiar from other sciences: temperature can't be directly observed but has to be inferred through the behaviour of thermometers. Grammaticality likewise has to be inferred from several measures, with acceptability ratings as one imperfect window. The same caution applies to language-model probabilities: raw string probability by itself doesn't separate grammatical from ungrammatical strings, and minimal-pair probability contrasts are informative only when the intended message is controlled (J. Hu et al., 2026).

The formal counterpart of this two-layer picture, including what it implies for interpreting the factor analyses that are common in experimental syntax, appears in the formal core (§3.7, §3.8): because such analyses target the structure of the subjective feeling, not grammatical status directly, a factor counts as evidence about the grammar only when it also shows up in the independent indicators for licensing, compatibility, and confidence.

This also explains why satiation and gradient judgments don't automatically threaten categorical status. A construction can remain unlicensed in the community's system while triggering progressively weaker negative responses through familiarization. Conversely, subjective response can remain continuous even when the population licensing state is effectively categorical (§4.4). Treating ratings as evidence rather than definition keeps grammatical hypotheses empirically vulnerable without collapsing the linguistic system of a communicative situation into individual responses to it.

Misattribution effects. Sometimes the subjective feeling of ungrammaticality doesn't accurately reflect objective grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

When grammatical constructions feel ungrammatical. Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(20) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase and arrive at nonsense. But the sentence actually has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like (15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

When ungrammatical constructions escape detection. Conversely, objectively ungrammatical utterances may fail to trigger negative responses when the intended meaning is strong:

(21) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Under the relevant syntactic analysis, the comparison composes differently. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

(22) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.* (modified from Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 Mechanisms of grammatical change

The stability of form–value relations isn't static. Various motivations alter the evidence balance for new and established forms, explaining how grammatical systems evolve over time. These aren't a loose catalogue of causes: each is a route by which the evidence balance of §4.3 can change sign, so each predicts a direction of change (which forms should gain licensing and which should lose it) and feeds the actuation dynamics of §4.8.

Semantic motivations for reanalysis. Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

(23) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

(24) *It's going to rain.*

What once expressed physical trajectory now serves as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

Social motivations for accepting or rejecting forms. Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others; over time, those preferences shift what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

Situational licensing tracks changing social dynamics: forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

Structural motivations. Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

Processing constraints and memorability. Transmission tends to favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

Analogical extension. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(25) ? *I asked about what did she do.*

Though currently marginal, such forms appear to gain traction by analogy with main-clause interrogatives (*What did she do?*). If accepted, this analogical extension would represent structural motivation reshaping the grammar.

Iconic motivations. Forms are sometimes accepted precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(26) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Iconicity remains syntactically constrained, though: it occurs in pre-head modifiers but is ungrammatical as predicative complement:

(27) * *the problem was big big*

These motivations (semantic, social, structural, and iconic) can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), OVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 A formal core: grammaticality as conditioned stability

The framework separates two explanatory targets: (i) a state theory specifying the conditions under which a form–value pair is grammatical *for a population in a communicative situation* at time t; (ii) a dynamics module specifying how such states tend to arise and persist. The state theory specifies what makes a form grammatical at a time; the dynamics module explains how such grammatical states arise, persist, and change. In projectibility terms, the state theory specifies the PROFILE, while the dynamics module states its SUPPORT: what stabilizes the profile and, where the evidence warrants, what maintains it. Support claims come in grades that have to be earned separately. It is one thing to show that a profile stabilizes, another to show that a production–repair loop maintains it, and a still stronger thing to show that some process detects deviations and pushes the profile back. None of those stronger claims is inherited from the framework's name (Reynolds, 2026a, 2026b). In Woodward's interventionist terms, a stabilizer earns that status when an intervention on it would change the profile in an invariant, predictable way (Woodward, 2001, 2003).

3.1 Conditioning structure

Let $c \in \mathcal{C}$ be a CONDITIONING STATE, a construed communicative situation in Wiese's (2023) sense, together with whatever norm-centre is treated as relevant. Nothing here requires c to be externally given; interlocutors can misalign about c, and c can be learned, split, and renegotiated over time.

For many purposes it helps to think of c as a bundle that may include situational features (activity type, medium, footing), stable baselines tied to speaker ascription, and norm-orientation (discourse-community identification), but the formalism treats c abstractly as the conditioning variable that selects the relevant distribution of form–value relations.

§§1–2 variable	Formal object (§3)	The work it did
$\mathsf{map}(u, c)$	$1\left[\mathcal{A}(u, v) \neq \emptyset\right]$	structural viability: some covering assembly exists
$K(u, c)$, operator part	def(A): unification	operator-value coherence, now hard
$K(u, c)$, oddness part	plausibility term in R_i (§3.7)	content-oddness, now a feeling-channel cost
$C_t(u, c)$	posterior over $L_t(A, c)$	situational licensing, now mean *and* concentration
$\widetilde{G}_t = \mathsf{map} \cdot K \cdot C_t$	$G_t(f, v, c)$: posterior existence	graded status; the old product is a limiting case (§3.3.1)
$\tau(c)$	unchanged, §3.3.4	decision-theoretic read-out only

Table 3: Where the constitutive variables' work lives in the revised core. The prose architecture of §§1–2 is unchanged; the formal core carries it on different objects.

3.2 Objects

The core's unit is the construction, in the CxG sense of a learned form–value pairing at any grain (Fillmore et al., 1988; Goldberg, 1995). Formally, a CONSTRUCTION κ is a pair $(\tau_\kappa, \delta_\kappa)$: a form template with typed slots, and a value contribution mapping slot fillers to content together with a PARTIAL OPERATOR ASSIGNMENT ω_κ. Nothing here is level-specific: a construction can be the plural cell of a lexeme, a clause pattern, an intonational contour, or a code-mixed frame.

Operator assignments live in a space Ω of partial functions from a dimension set $O(c)$–clause type, polarity, tense–aspect anchoring, role allocation, scope, reference tracking, information-structural packaging–to finite value sets. Which dimensions belong to $O(c)$ is itself conventional: a contrast enters only when the community has conventionalized it as an operator contrast in the sense of §2, so the inventory is community-indexed, as the coherence constraints were before. Two partial assignments UNIFY, written $\omega \sqcup \omega'$, iff they agree on every dimension where both are defined; otherwise the join is undefined.

An ASSEMBLY A is a finite tree of constructions with all slots filled. Its yield is $\mathrm{form}(A)$; its operator assignment is $\omega(A) = \bigsqcup_{\kappa \in A} \omega_\kappa$, defined iff all contributions unify; its value is $\mathrm{val}(A)$. Three predicates over assemblies carry the old variables' work:

$$\begin{aligned} \mathrm{def}(A) &= 1\left[\omega(A) \text{ is defined}\right] && \text{(compatibility)} && (28)\\ \mathrm{sat}(A,c) &= 1\left[\omega(A) \text{ is total on } \mathrm{OBL}(c,A)\right] && \text{(saturation)} && (29)\\ L_t(A,c) &= \textstyle\bigwedge_{\kappa \in A} z_t(\kappa, c) && \text{(licensing)} && (30) \end{aligned}$$

Compatibility is hard: there is no weighting an operator clash away. Saturation refers to the obligatory dimensions $\mathrm{OBL}(c, A)$, which are not stipulated per language but derived from the dynamics. A dimension is obligatory exactly when zero-marking assemblies have been preempted to near-zero licensing in every niche where it is at issue (§4.5). English progressive marking and Turkish evidential marking are the worked cases there.

Licensing is latent. Each construction–situation pair carries an inclusion variable $z_t(\kappa, c) \in \{0, 1\}$: whether κ belongs to the community repertoire at c. Nobody observes z; speakers infer it from tokens, structured omissions, and repair, and the learner's posterior is the node-level Beta posterior of §4.3, with mean $C_t(\kappa, c)$ and concentration $\nu_t(\kappa, c)$. The individual state $\pi_{i,t}$ and the population distribution over it are retained; what changes is only what the posterior now feeds.

Partial pooling survives restated. A novel token such as *She texted him the address* need not have been encountered: the assemblies covering it are built from nodes–ditransitive transfer, pronominal object order, the relevant lexical-class relations–each with its own posterior, and the token inherits its status through them. What the learner updates is constructional regions, not memorized sentence types, so first-heard sentences are not predicted to have zero licensing.

3.3 Grammatical status as a conditioned state property

3.3.1 Status as posterior existence

For a form f, value v, and conditioning state c, let $\mathcal{A}(f, v)$ be the set of assemblies with yield f and value v. Grammatical status is the posterior probability that a licensed route to the form–value pair exists:

$$G_t(f, v, c) = P\Big(\, \exists A \in \mathcal{A}(f, v) :\ \mathrm{def}(A) \wedge \mathrm{sat}(A, c) \wedge L_t(A, c) \ \Big|\ \mathcal{D}_t \Big), \tag{31}$$

where $\mathcal{D}_t$ is the evidence to date: tokens, structured omissions, repairs, avoidance, and situation cues. Status is a property of a population–situation pair, read through a learner's or an analyst's posterior; the population state it estimates is the distribution of node posteriors induced by c, exactly as before. Where value is recoverable from context, I write $G_t(u, c)$ for readability.

Two remarks keep (31) honest. First, finiteness: for a finite yield and a finite, non-empty-recursive construction inventory (no cycles of null-yield expansions–the standard offline-parsability condition), $\mathcal{A}(f, v)$ is finite, so the existential is a finite disjunction. No tractability escape hatch is needed.

Second, the relation to the previous formulation. In the high-precision limit, where every node posterior has concentrated, (31) has the same limiting classification as a weakest-link rule over the best assembly: a licensed, defined, saturated assembly exists or it does not. And under node-independence with a single dominant assembly A^*, the probability factorizes,

$$G_t(f, v, c) \approx 1\left[\mathrm{def}(A^*)\right] \cdot 1\left[\mathrm{sat}(A^*, c)\right] \cdot \prod_{\kappa \in A^*} C_t(\kappa, c). \tag{32}$$

This is why the earlier three-factor product worked as well as it did: it is the single-assembly, independent, high-precision special case of (31), with structural viability as non-emptiness and hard compatibility folded into def. Away from that limit, (31) replaces weakest-link scoring with a posterior over assembly existence, and the uncertainty it carries is the point.

Where the previous draft also folded content-oddness into coherence–the concentration of construal for cases like *colorless green ideas*–that work now lives in the feeling model's cost vector (§3.7). A lexically bizarre but operator-coherent construal leaves G_t untouched and shows up as a plausibility cost in R_i: the sentence is grammatical and odd, which is the datum.

3.3.2 Projectible use of the posterior state

G_t is not bookkeeping after the real explanation has happened. It is the macro-state over which grammaticality projects–and it is now two-dimensional by construction, so Figure 2 depicts the formal object itself rather than a commentary on a scalar that discards half of it.

Some generalizations run over the posterior mean. Production and repair behaviour at $t+1$ depend partly on $\mathbb{E}[G_t]$:

$$\Pr(\text{produce } u \mid n, c, t) \propto \mathbb{E}\big[G_t(u, c)\big]\ \rho_t^{\star}(u \mid n, c), \tag{33}$$

$$\Pr(\text{repair } u \mid c, t) = r_0 + r_1\big(1 - \mathbb{E}\big[G_t(u, c)\big]\big), \qquad r_1 > 0. \tag{34}$$

Other generalizations run over the concentration, and they are the diagnostic ones. Two form–value pairs can share a low mean while differing in how much evidence backs it, and the update dynamics make them behave differently under new exposure: a diffuse posterior moves with framing and familiarization; a concentrated one does not. The status regions are:

- licensed: $\mathbb{E}[G_t] \approx 1$, $\mathrm{Var}(G_t) \approx 0$;
- excluded: $\mathbb{E}[G_t] \approx 0$, $\mathrm{Var}(G_t) \approx 0$–confident rejection, whether by structural failure or by concentrated non-licensing;
- unsettled: $\mathrm{Var}(G_t)$ high, in two subtypes–*starved* (low concentration from a sparse opportunity set, as with independent relative *whose*) and *divided* (low concentration despite dense opportunities, from conflicting or absorbed evidence flows, as with defective cells, §4.6).

The projectible generalizations are as before–satiation profiles, framing sensitivity, transmission fidelity, repair rates, actuation–but they now run over a state that carries the dimension they turn on. Left-branch extraction and *whose* approach similar means and part on concentration; *amn't* parts from left-branch extraction the same way. The feedback loop closes as before: production and repair generate the later evidence streams, individual behaviour updates population licensing, and the population state projects back into future production and repair.

3.3.3 Categorical talk as posterior concentration

Communities often treat grammaticality as a membership fact: a relationship counts as an available resource in c or it does not. On the present account, that talk is licensed exactly where the posterior has concentrated–where the licensing state sits near an attractor and $\mathrm{Var}(G_t) \approx 0$. Categoricality is a property of the state, not a threshold imposed on it; §4.4 derives the bimodality that makes the

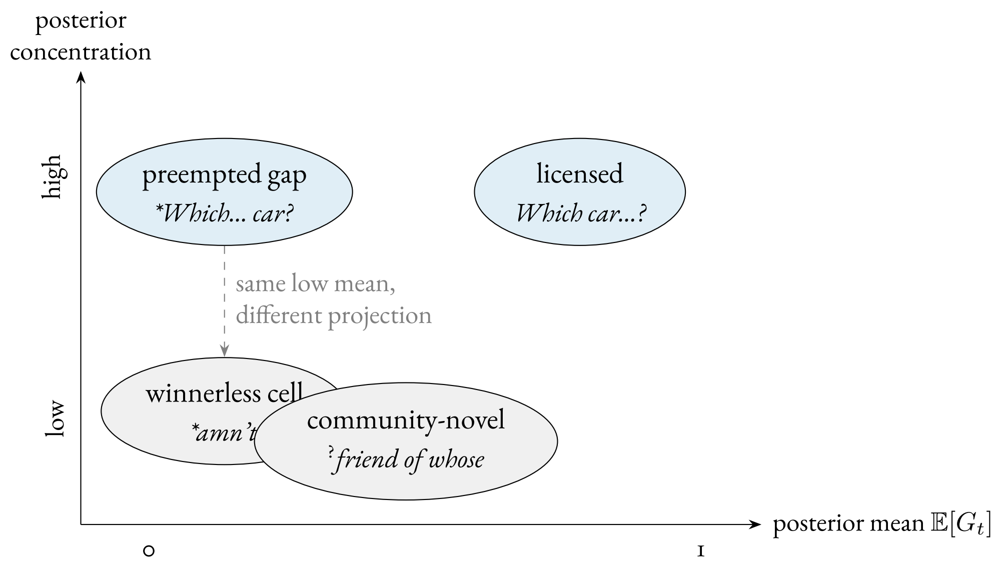

Figure 2: The posterior status state has two dimensions. Preempted gaps (dense preemption, posterior concentrated near zero) and winnerless cells (transparent candidates, low mean but diffuse evidence) share a low mean and part only on concentration, the axis that predicts satiation, framing sensitivity, and actuation.

categorical regime the typical one for closed, high-opportunity, update-configuring contrasts. In the contested middle, categorical talk is predicted to be unstable and stakes-sensitive–which is the datum, not a bug.

3.3.4 A decision-theoretic read-out

The threshold survives, demoted to where it belongs. A judge in c choosing between labels $\ell \in \{\text{IN-REPERTOIRE}, \text{NOT-IN-REPERTOIRE}\}$ under asymmetric losses $L_{\text{fa}}(c)$ and $L_{\text{fr}}(c)$ optimally accepts iff

$$\mathbb{E}\big[G_t(u,c)\big] \geq \tau(c) = \frac{L_{\text{fa}}(c)}{L_{\text{fa}}(c) + L_{\text{fr}}(c)}. \tag{35}$$

$\tau(c)$ does no constitutive work: it is a decision criterion induced by payoff structure, strict in high-stakes institutional contexts, permissive in in-group ones. Where the posterior is bimodal, the read-out is insensitive to the exact value of $\tau(c)$; in the middle, judgments should be unstable and stakes-sensitive. The threshold-discontinuity test of §6.1 is untouched, including its distinctive prediction that manipulating stakes moves the location of the discontinuity.

Two clarifications keep the categorical read-out from overclaiming. First, calling the status "objective" means population-level and convention-constituted, not judge-independent in any stronger sense. The status stands apart from any individual's response the way facts about word meaning or currency do; it is worldly in that, given the community's practices, the projected relations are not made available by the analyst's decision to group cases together (Reynolds, 2026a). CONVENTIONAL STATUS would be an equally accurate name. Second, the threshold's arbitrariness is confined by the dynamics: where licensing posteriors are bimodal, no point estimate of $\tau(c)$ is needed; in the contested middle, $\tau(c)$ is set by independent stakes cues.

3.4 What the existential does and doesn't do

The place of the old mapping predicate is taken by the non-emptiness of $\mathcal{A}(f, v)$: genuine analyzability failure is the case where no assembly covers the string at all, and then $G_t = 0$ exactly, from an existential over the empty set. This remains the only failure mode that is categorical in the representational sense.

Analyzability is relative to the constructional inventory, stored wholes included, not to productive composition alone. Syntactically anomalous idioms make the difference visible: *by and large*, *far be it from me*, and *come Monday* have no analysis by the productive syntax of English, but they are fully conventional (Fillmore et al., 1988). They are covered by their own stored constructions–coverage by some node, productive or memorized, is what analyzability means here. The cost is the same candour as before: non-emptiness is not pre-conventional. What it contributes is a different kind of failure (no covering assembly at all), not a convention-free one.

The residual clause is gone. The previous draft reserved a case for "an independent constraint on possible analysis"; the revised core has no work for it. A string with no covering assembly gets $G_t = 0$ from the empty existential; a preempted gap gets $G_t \approx 0$ from a covering assembly licensed to zero. Both are excluded, and they differ in *why*–no assembly versus licensed-to-zero assembly–a difference with an empirical signature: the first fails under every construal and resists all framing; the second has a transparent construal, resists framing by concentration (§3.3.2), and is predicted to be re-openable only by the actuation route of §4.8. The constitutive core loses the escape hatch and gives up nothing.

Because the existential is over value-matched assemblies, homophonous strings can split by construal. A form with one clashing parse and one coherent parse is grammatical on the coherent value and excluded on the clashing value; ambiguity does not let a failed assembly contaminate a successful one.

3.4.1 Assigning a failure

The three-way diagnostic of the previous draft becomes a four-step procedure, applied before consulting acceptability data and checkable against the independent indicators of §3.6 at each step.

Ask first whether any assembly covers u: if none does, the failure is structural (**Can the have running*). If a covering assembly exists, ask whether its operator contributions unify: if every covering assembly sets some dimension twice incompatibly, the failure is compatibility (**I've finished it yesterday*–two values on the reference-interval dimension; and the information-structural clash in Cuneo and Goldberg's island example is the same formal object on a different dimension (Cuneo & Goldberg, 2022)). If the contributions unify, ask whether the obligatory dimensions are saturated: if not, the failure is saturation (unmarked progressive in an ongoing-at-issue English niche). Only then does licensing arbitrate: a defined, saturated assembly whose pivotal node the population withholds is a licensing failure (**I have 25 years* for age–the frame is coherent, the value transparent, and the age-predication node sits at $C_t \approx 0$ under preemption by the BE-frame).

Because the dimension inventory $O(c)$ is itself conventionalized, the ordering point of the previous draft survives: what counts as a compatibility failure in c depends on which contrasts the community has conventionalized, so the steps are ordered, not independent–and each step's verdict is checkable against evidence other than the judgment it explains, which is what keeps the procedure from reproducing the circularity of a pure community-verdict account.

3.5 Interlocutor misalignment about conditioning

Because c is construed, interlocutors can disagree about which conditioning state is in force. Let a hearer h maintain a posterior $q_h(c \mid e)$ over conditioning states given cues e (situational cues, ascription cues, stance cues, etc.). Then the hearer's expected status is

$$\hat{G}_t^{(h)}(u \mid e) = \sum_{c \in \mathcal{C}} G_t(u, c)\, q_h(c \mid e).$$

3.6 Identifying the conditioning state before prediction

The conditioning state c is not a free parameter to be selected after a judgment is observed. For empirical work, c has to be fixed before predicting ratings or repair behaviour. The posterior $q_h(c \mid e)$ is the identification device: it is estimated from non-judgment cues such as activity type, medium, participant roles, speaker ascription, register cues, and overt norm orientation. A form like *might could* is not rescued by choosing a sympathetic norm-centre after the fact; the model predicts different judgments only when the independent cues make that norm-centre probable.

This also blocks a competence/performance redux. C_t, the population licensing state, is identified by corpus-rate-per-opportunity, elicited production, and repair behaviour, not by acceptability ratings alone. F, the subjective-feeling variable, is identified by ratings, online measures, and self-reports. The confidence read-out Φ (§3.7) is identified by test–retest stability, framing lability, and explicit confidence ratings, none of which enter the estimation of the licensing posterior itself. The model is falsified where these dissociate in the wrong direction: if satiation changes later corpus rates for a purportedly preempted gap, if repair behaviour tracks prescriptive dissonance rather than licensing, or if high-opportunity preempted forms behave like low-evidence uncertainties, the state assignment is wrong.

Reassignment is not free, though. Localizing a failed prediction to a narrower conditioning state is warranted only when that state is specifiable independently of the surviving cases, by activity type, medium, participant roles, or norm-orientation fixed before the failures were seen (Reynolds, 2026a). A partition of $\mathcal{C}$ redrawn around exactly the successes redescribes failure as success; it does not rescue the model. Confirmatory studies should register the conditioning partition and the estimation protocol for $\rho_t^\star$ (e.g. forced-choice norming over the competitor set, estimated before the licensing data is touched). The corpus checks in this paper are illustrative probes, not registered tests. Under that discipline, repeated post-hoc reassignment across future studies counts against the framework itself, not just against particular state assignments.

Two identification points deserve to be explicit, because a corpus rate estimates the product of licensing and counterfactual choice (§4.2), not licensing alone. First, $\rho_t^\star$ has to be estimated from evidence that does not include the licensing data it later interprets. A forced-choice norming task supplies it: present the niche n and the competitor set $\mathcal{V}_{n,t}^\star$ to a separate sample and record how often each variant is chosen when stipulated to be available. That yields $\rho_t^\star$ before any corpus rate for u is consulted, so a low rate can be diagnosed as non-licensing ($C_t \approx 0$, $\rho_t^\star$ high) rather than licensed-but-dispreferred selection (C_t high, $\rho_t^\star \approx 0$). Second, $\tau(c)$ is not fit to the judgments it predicts. Where the licensing posterior is bimodal (§4.4), G_t is insensitive to the exact threshold, so no point estimate of $\tau(c)$ is needed; in the contested middle, $\tau(c)$ is set by independent stakes cues (institutional versus in-group footing), and the movable-cutoff prediction of §3.3.4 tests it directly: manipulate the stakes and the location of any discontinuity should move, which cannot happen if $\tau(c)$ were merely a curve fitted to the responses.

3.7 The feeling of (un)grammaticality

Speakers register putative violations through a metacognitive signal that can diverge from status; the two are identified by different evidence streams (§3.6). The revision makes the observable phenomenology a pair: a signed anomaly signal and a confidence read-out. The pair is what lets the model derive, rather than assert, the contrast between confident rejection and ineffability.

Let $\hat{G}^{(h)}_{i,t}(u \mid e)$ be hearer i's expected status under situation uncertainty (§3.5). The anomaly signal accumulates low expected status, implementation costs, and ideological overlays:

$$A_{i,t}(u,c) = \alpha\big(1 - \hat{G}^{(h)}_{i,t}(u \mid e)\big) + \gamma^\top R_i(u,c) + \delta\, D_i(u,c) + \varepsilon_i, \tag{36}$$

$$F_{i,t}(u,c) = -\frac{\max\{A_{i,t},0\}}{1+\max\{A_{i,t},0\}} \in (-1,0]. \tag{37}$$

The bounded link and the floor at zero are as before: there is no positive feeling of grammaticality, only the negative signal registered on deviation. D_i is prescriptive dissonance. The cost vector R_i keeps its processing components–the locality term $L(u)$ with its saturating form, interference, garden-path reanalysis, surprisal–and gains a plausibility component: the implausibility of the best construal given world knowledge and lexical expectations. That component absorbs the retired coherence machinery's oddness work. *Colorless green ideas sleep furiously* is defined, saturated, and licensed node by node, so $G_t \approx 1$; its strangeness enters here, as a construal-plausibility cost, and the sentence comes out grammatical and odd–the datum, with one less primitive.

The locality component can still be written explicitly:

$$L(u) = \sum_{d \in \mathcal{D}(u)} \ell\big(|d|\big), \qquad \ell(k) = \begin{cases} k, & k \le K_{\text{sat}}, \\ K_{\text{sat}} + \beta\,(k - K_{\text{sat}})^{\eta}, & k > K_{\text{sat}}. \end{cases}$$

Here $|d|$ is the length of dependency d (in intervening discourse referents or words), K_{sat} is a saturation threshold beyond which additional distance contributes sublinearly, and β, η (with $0 < \eta < 1$) control the rate of that sublinear growth.

The second read-out is confidence. Let $\nu_i(u,c)$ be the concentration of i's posterior over the *pivotal condition*–the conjunct in (31) on which the token's status turns. For structural and compatibility failures the pivotal condition is a fact about the assembly set, so once the relevant inventory is fixed the posterior over it is treated as effectively degenerate: confidence is high by modelling convention, not by a separate learning theorem. For licensing-pivotal tokens, ν_i is the Beta concentration of the pivotal node's posterior (§4.3). Define

$$\Phi_{i,t}(u,c) = \frac{\nu_i(u,c)}{\nu_i(u,c)+\nu_0} \in [0,1), \tag{38}$$

with ν_0 a scaling constant. The pair (F, Φ) is the observable phenomenology, and the predictions separate:

- *No assembly* (**Can the have running*): F strongly negative, $\Phi \approx 1$, under every construal.
- *Compatibility failure* (**I've finished it yesterday*): F strongly negative, $\Phi \approx 1$; entrenchment of the parts cannot repair it, because unification is hard.
- *Preempted gap* (left-branch extraction): F strongly negative, Φ high–"that's impossible"; framing-resistant.
- *Defective cell* (**amn't*, *pobedit'* 1SG): F negative, Φ low–"I don't know how to say it"; framing-movable.
- *Community-novel* (independent relative *whose*): F weakly negative, Φ low; framing-movable, with surprisal adding R-channel noise.
- *Processing overload* (centre embedding): F negative via R_i, Φ moderate; satiation lowers the R contribution, moving F toward zero without touching the licensing posterior.

Optionally, the model can carry an attribution variable: on anomaly detection the hearer infers a source $z \in$ {UNLICENSED, OTHER-COMMUNITY, SPEAKER ERROR, NOISE, OWN OVERLOAD}, with the posterior over z shaped by Φ and by the same ascription machinery as $q_h(c \mid e)$. That derives, rather than stipulates, why learner errors and cross-dialect tokens fail to trigger the ungrammaticality feeling–they are attributed to OTHER-COMMUNITY or ERROR–and it gives the misattribution effects of §2.4 a formal home. The attribution layer is a measurement refinement, not part of status.

One falsifier is worth stating here rather than in §6.1: if measured confidence fails to dissociate left-branch extraction from *amn't* at matched mean ratings, the concentration dimension is doing no work and the revision to (31) loses its main empirical payoff.

3.8 Measurement: four observation channels

Status is latent; it is estimated from converging observation channels, none of which defines it. The revision makes the channels explicit:

$P(\text{production} \mid z, n, c)$	choice among assemblies in a niche (§4.2)	(39)
$P(\text{repair} \mid z, \Delta, D, c)$	sensitive to operator mis-setting (§4.3)	(40)
$P(\text{judgment} \mid G_t, R, D, \nu)$	the (F, Φ) read-out of §3.7	(41)
$P(c \mid \text{cues})$	situation inference, $q_h(c \mid e)$ (§3.5)	(42)

Production is choice among assemblies, not a direct read of licensing: a corpus rate estimates the product of licensing and counterfactual choice, exactly as in §4.2, and the identification discipline of §3.6 (norming $\rho_t^\star$ on independent data before the licensing data is touched) carries over unchanged. Repair enters with the sign predicted in §3.3.2 and, in the revised dynamics, with the Δ-sensitivity of the repair stream (§4.3). Judgment is a metacognitive read-out, not status; the confidence channel Φ is identified by test–retest stability, framing lability, and explicit confidence ratings. Situation is inferred from cues, not stipulated after the fact.

The factor-analysis discussion survives with one amendment. Factor analyses of acceptability ratings target the structure of the judgment channel; a factor counts as evidence about the grammar only when it also shows up in the independent indicators for licensing and compatibility. The amendment: judgment studies should now report dispersion and test–retest structure alongside means, because the model's sharpest projections–satiation rank order, framing lability, the gap/cell contrast–run over the concentration that means discard.

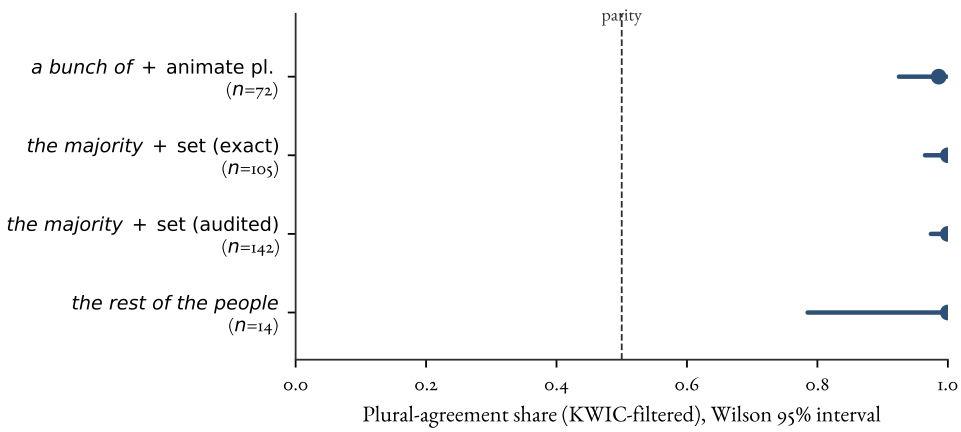

Figure 3: Corpus-rate-per-opportunity as an estimator of licensing for a contested agreement contrast. Plural-agreement share (dot) with Wilson 95% interval (bar) for collective-partitive subjects in COCA, after KWIC filtering to subject-position tokens; the dashed line marks parity (no directional preference). Data: `agr-coca-projection` probe (*bunch*, *majority*/*minority*, and partitive cells). Notional plural dominates in every cell, and the interval widens with the smaller denominators rather than drifting toward parity.

The corpus-rate-per-opportunity indicator is concrete. Figure 3 shows one worked case: agreement on collective-partitive subjects (*a bunch of people*, *the majority*), where surface-head number (singular) and notional number (plural) diverge. Counting only subject-position tokens after KWIC filtering (so non-subject false positives are removed) gives a plural-agreement share with a Wilson interval for each cell. The shares sit far above parity in every cell, with the interval width tracking the token count rather than the direction. This gives a licensing estimate for a contested operator contrast read off usage per opportunity instead of ratings: a latent licensing posterior recovered from a converging indicator, with its uncertainty made explicit.

Table 4 makes the figure auditable: exact filtered cell counts, so the estimate can be checked rather than taken on trust.

Table 4: Filtered agreement cells behind Figure 3. Counts are KWIC-filtered subject-position tokens in COCA; "pl." and "sg." are plural- and singular-agreement targets. The surface-head baseline predicts singular; the observed share is plural.

Cell	pl.	sg.	N	Plural share (Wilson 95%)
a bunch of + animate pl.	71	1	72	0.986 (0.925–0.998)
the majority + set (exact)	105	0	105	1.000 (0.965–1.000)
the majority + set (audited)	142	0	142	1.000 (0.974–1.000)
the rest of the people	14	0	14	1.000 (0.785–1.000)

3.9 Worked examples: deriving status from the posterior

The canonical examples of §2.2, recomputed, with the winnerless cell added. The previous draft asserted its phenomenology and the formal core never touched it; now it is derived.

1. **Can the have running* (nonsense). $\mathcal{A}(f, v) = \emptyset$: no covering assembly. $G_t = 0$ exactly; $\Phi \approx 1$ under every construal. Status: ungrammatical, structurally.
2. *Colorless green ideas sleep furiously*. A covering assembly is defined, saturated, and licensed node by node: $G_t \approx 1$. The strangeness is a construal-plausibility cost in R_i: F negative, status grammatical. Content-odd, not ill-formed.
3. **I've finished it yesterday* (compatibility failure). Covering assemblies exist, but every one sets the reference-interval dimension twice incompatibly–the perfect's contribution and the deictic adverb's do not unify–so $\mathrm{def}(A) = 0$ throughout. $G_t = 0$; $\Phi \approx 1$. Entrenchment of the component constructions cannot repair it: the clash is outside the licensing terms altogether.
4. **I have 25 years* (licensing failure). Defined and saturated; the value is transparent. The pivotal age-predication node sits at $C_t \approx 0$ with high ν, preempted by the BE-frame in a dense niche. $\mathbb{E}[G_t] \approx 0$, Φ high. Status: ungrammatical, by concentrated non-licensing.
5. *?friend of whose* (community-novel). Defined and saturated; the niche is vanishingly sparse, so the pivotal node's posterior sits near its prior with low ν. $\mathbb{E}[G_t]$ intermediate, $\mathrm{Var}(G_t)$ high, Φ low. Status: unsettled (starved); framing-movable, with surprisal adding R-channel noise.
6. **Which did you buy car?* (preempted gap). Defined and saturated; the dedicated discontinuity node–the only activated node covering the determiner–head discontinuity–absorbs preemption mass from every fronting opportunity, and its posterior is driven to zero and concentrates there (§4.3). $\mathbb{E}[G_t] \approx 0$, $\Phi \approx 1$. Status: excluded; framing-resistant, non-satiating.
7. **amn't, pobedit'* 1SG (winnerless cell). The niche is dense and candidate assemblies are transparent, but no candidate accumulates positive evidence–the outside option absorbs production, and because avoidance has low attributability, candidates accumulate little confident negative evidence either (§4.6). Every candidate's pivotal node has low mean and low ν. $\mathbb{E}[G_t]$ low, $\mathrm{Var}(G_t)$ high, Φ low. Status: unsettled (divided); phenomenology ineffability, not rejection–same mean as the preempted gap, different concentration, different feeling.
8. *The bread the baker the apprentice helped made is delicious* (processing overload). G_t high; the locality term in R_i is large. F negative, Φ moderate; satiation and instruction move the R contribution, not the licensing posterior. Status: grammatical, transiently ill-felt.

Figure 4 shows the low-opportunity versus dense-preemption contrast that underwrites several of these classifications.

4 Dynamics: why the state stabilizes

The dynamics module explains trajectories of construction-level licensing posteriors, and in principle refinements of the conditioning partition itself. It does not define grammaticality.

4.1 Niches, competitors, and opportunity sets

Let n index a CONSTRUCTIONAL NICHE (a communicative job), and let $\mathcal{V}^{\star}_{n,t}$ be the opportunity set: the observed, licensed, and analogically generated variants that could plausibly do that job in some conditioning states. Let $N_t(n, c)$ be the number of opportunities for niche n in conditioning state c over some time window, and $k_t(v, n, c)$ the observed count of variant $v \in \mathcal{V}^{\star}_{n,t}$.

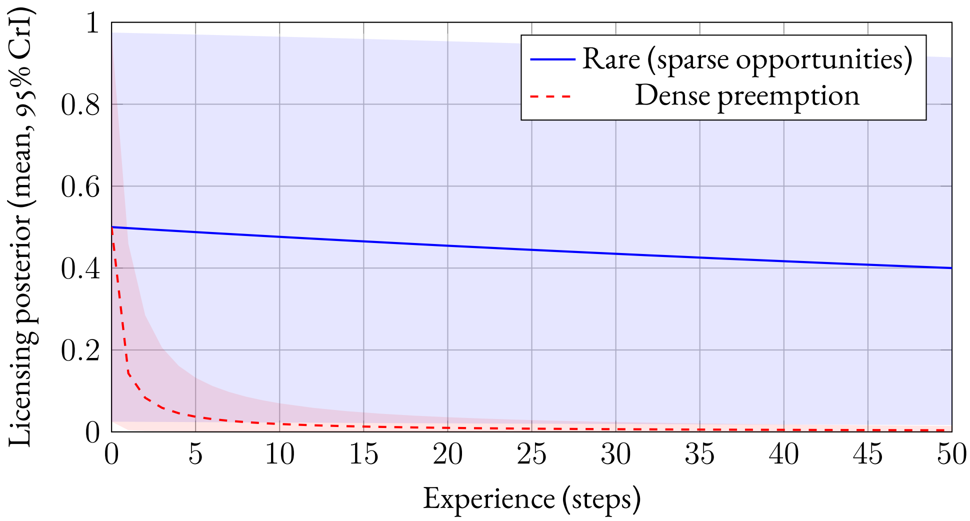

Figure 4: Licensing posteriors under preemption, from a Beta$(1, 1)$ prior with per-step effective preemption mass 0.01 (rare, blue) versus 5 (dense preemption, red); lines are posterior means, shaded regions 95% credible intervals. The rare construction stays near its prior with a credible interval spanning nearly the whole unit interval, while the dense-preemption posterior both falls and concentrates. The projections in §3.3.2 that turn on evidence structure–satiation, framing sensitivity, actuation–run over this dispersion difference, which means alone do not encode.

The star matters. Preempted gaps and innovations often involve variants with no positive tokens. Their counterfactual utilities are still defined because the hierarchy $\mathcal{H}$ generates analogical candidates. English learners can represent a left-branch extraction candidate by combining interrogative fronting, modifier–noun packaging, and question formation, even if no one licenses that candidate as an English resource. The same machinery lets a novel but ordinary sentence inherit coverage from licensed higher-grain constructions rather than being preempted merely because that exact sentence type is new.

4.2 Usage as licensing × choice among candidates

Separate LICENSING from SELECTION. A variant can be licensed but rarely chosen.

Let $\rho_t^\star(v \mid n, c) \in [0, 1]$ be the counterfactual probability of choosing v if it were available in niche n under c. A flexible choice model is a softmax over utilities:

$$\rho_t^\star(v \mid n, c) = \frac{e^{U_t(v;n,c)}}{\sum_{v' \in \mathcal{V}_{n,t}^\star} e^{U_t(v';n,c)}}, \qquad U_t(v; n, c) = \boldsymbol{\theta}^\top \mathrm{f}(v; n, c).$$

At the population level, the expected usage rate $\pi_t(v \mid n, c) \approx C_t(v, c) \cdot \rho_t^\star(v \mid n, c)$. The counterfactual choice term can be high even when $C_t(v, c)$ is near zero; that is exactly the condition under which absence becomes informative.

4.3 Dynamics: evidence streams, weighted omissions, and repair

The learner represents each construction–situation pair (κ, c) with a Beta posterior over the latent inclusion variable $z_t(\kappa, c)$ (§3.2). The update runs on discounted counts: with memory discount

$\delta_m \in (0, 1]$, an evidence item at occasion t_i contributes $\delta_m^{t-t_i}$ of its weight at time t. Setting $\delta_m = 1$ recovers perfect retention; $\delta_m < 1$ gives bounded evidence memory, and it is what the destabilization result in §4.7 turns on.

The streams are positive evidence s_t, error evidence e_t, and preemption mass p_t, distributed over activated nodes by the coverage weights:

$$a_{g,t+1} = \delta_m a_{g,t} + \lambda_g(u, c) s_t(u, c), \tag{43}$$

$$b_{g,t+1} = \delta_m b_{g,t} + \lambda_g(u, c)\big\{e_t(u, c) + p_t(u, c)\big\}. \tag{44}$$

The node-level mean and concentration are

$$C_{t+1}(g, c) = \frac{a_{g,t+1}}{a_{g,t+1} + b_{g,t+1}}, \qquad \nu_{t+1}(g, c) = a_{g,t+1} + b_{g,t+1}. \tag{45}$$

The ascription weighting is unchanged. Each token's contribution is scaled by the hearer's posterior that it is a competent realization of the norms of c. This is why child and learner errors barely move adult posteriors, and why koinéization falls out as a shift in ascription weights rather than a separate mechanism.

Weighted omissions. The previous draft treated absence as evidential in proportion to the counterfactual choice probability: $p_t = \sum_j \rho_t^\star(u \mid n(u), c)$ over niche opportunities. That was right as far as it went, but it went silent on occasions where the niche arose and nothing in the candidate set was produced–periphrasis, avoidance, repair-in-progress. Counting those as negative evidence would make every defective cell collapse into a preempted gap. The fix falls out of inverting the production model.

Consider an occasion i where niche n arose and some $v_i \neq u$ was produced. The evidential question is how strongly that choice tells against u's membership in the repertoire. Under the production model, if u is licensed it is chosen with probability $\rho_t^\star(u \mid n, c)$, and the remaining probability renormalizes over the alternatives; if u is unlicensed, the alternatives carry the full mass. The per-occasion evidence weight for non-licensing is therefore

$$q_i = 1 - \frac{P(v_i \mid z_u = 1, n, c)}{P(v_i \mid z_u = 0, n, c)}, \tag{46}$$

the attributability of the omission.

Two regimes matter. When v_i is an in-repertoire competitor, the renormalization gives $q_i = \rho_t^\star(u \mid n, c)$: the old preemption mass is recovered exactly, now as the first-order total of per-occasion Bayes factors rather than a definition. When v_i is the outside option–periphrasis or avoidance, available at fixed utility under both hypotheses–its choice probability is nearly the same whether u is licensed or not, provided u's counterfactual utility is low or the production rule assigns a face cost to weakly licensed forms. The condition is independently estimable from the same production data that estimates $\rho_t^\star$: speakers who avoid an exposed candidate despite understanding the intended value are treating production itself as costly. Then $q_i \approx 0$: taking the outside option is nearly uninformative about u.

A second weight scales for niche identification: r_i is the posterior probability that occasion i instantiated a job u would have done at all, with the intended value recoverable under u. The generalized preemption mass is then

$$p_t(u, c) = \sum_{i\text{: omissions}} r_i q_i \, \delta_m^{t-t_i}, \tag{47}$$

which reduces to the previous draft's $N_t \cdot \rho_t^\star$ when every omission is a competitor choice in a clearly identified niche with $\delta_m = 1$. Neither weight is a free parameter: both are read off the production model and the niche posterior. This is what keeps periphrasis from counting as confident negative evidence, and so what separates the defective cell's low-mean/low-concentration state from the preempted gap's low-mean/high-concentration one (§4.6, §3.9).

The repair stream: a candidate controller. The paper's support claims come in grades: stabilization, maintenance, control (§5). The dynamics so far deliver the first two. The third needs a process that registers departures from the licensed profile and pushes the population back–and conversational repair is the obvious candidate, with a formal home.

Let $\Delta(d)$ be the expected common-ground divergence from mis-setting operator dimension d: the quantitative face of "update-configuring" in the working test of §2. When a token mis-sets d, other-initiated repair occurs with probability $r(\Delta, \iota)$, increasing in Δ and modulated by footing ι. Each repair event feeds the evidence streams: error evidence e_t for the deviant node, positive evidence s_t for the repaired-to competitor. Over a window with $N_t(n, c)$ opportunities, the corrective flow against a deviant convention in niche n is

$$\phi_{\text{rep}}(n) = N_t(n, c) \cdot P(\text{mis-set}) \cdot r(\Delta, \iota), \tag{48}$$

a restoring force whose magnitude scales with the product of opportunity and divergence.

In the mean-field dynamics (§4.4) this adds a term pushing C_t back toward the community's attractor after perturbation. That is Woodward's interventionist signature: the repair rate earns controller status if an intervention on it changes the profile's return-to-attractor behaviour in an invariant, predictable way (Woodward, 2001, 2003). The claim stays graded: what the term establishes is a testable candidate controller, not a demonstration that grammar is controlled.

Two commitments come with the term. First, the operator/style boundary becomes a parameter statement: policing intensity is predicted to scale with $N_t \cdot \Delta$, with opportunity estimable from opportunity-annotated corpora and Δ from repair-initiation rates in conversation-analytic corpora regressed on annotated update divergence. The §2 falsifier–a closed, high-opportunity, non-configuring contrast policed like grammar–now has a quantitative form: such a contrast has $\Delta \approx 0$, so its corrective flow collapses to a social-sanction baseline and its licensing distribution should stay gradient, policed like style. Second, if repair rates track social indexing rather than Δ, the corrective term collapses into sociolinguistic policing and the controller claim fails cleanly. Blythe and Croft's utterance-selection dynamics remain the $r \equiv 0$ special case, so the import is conservative (Blythe & Croft, 2012).

Grain asymmetry and partial pooling. The two kinds of evidence attach at different grains. Positive evidence distributes over activated nodes by the coverage weights: a heard token confirms every construction that covers it. Error and preemption mass instead concentrate on the most specific node that individuates the candidate within its niche, because that node is what distinctively predicts the missing tokens. In Bayesian terms, the likelihood penalty for structured non-occurrence falls on the hypothesis that wastes probability mass on sentence types never observed; likelihood is a quantitative measure of the weight of implicit negative evidence (Perfors et al., 2011).

Partial pooling raises an objection. Token-level licensing is assembled from constructional nodes, so left-branch extraction activates the well-licensed nodes for interrogative fronting, question formation, and modifier–noun packaging. The sum can look as though it should hand the candidate substantial licensing.

The coverage weights block that result. A node contributes to a token's pooled licensing only for the properties it actually sanctions, and those parent nodes sanction only *contiguous* realizations. None of them covers the determiner–head discontinuity that is the distinctive property of **Which did you buy car?*; producing that token requires a node that licenses a determiner–head arc crossing the verb, and no contiguity-preserving node in $\mathcal{H}$ generates one (Reynolds, 2026d).

The parents keep their high licensing, each confirmed by its own abundant contiguous tokens, and contribute none of it to the discontinuous token through the pooling sum. The dedicated left-branch-extraction node, the only activated node that covers the discontinuity, carries near-gating weight for it and absorbs the preemption mass from every fronting opportunity where a contiguity-preserving competitor wins. The dedicated node is required by the derivation, not gerrymandered to force the result.

This asymmetry also answers an acquisition worry. If token-level licensing were just pooled parent licensing, learners should pass through a stage of treating left-branch extraction as live. They do not need to: the candidate is generated by analogy, so it is on the table with non-zero prior support, but its dedicated node has no scaffolding of its own, fronting opportunities are dense in the input, and each one adds preemption mass. A companion corpus study quantifies the evidential situation: across 1,228 fronting opportunities in a sixteen-corpus Universal Dependencies sample, zero genuine determiner–head discontinuities occur (95% upper bound 0.24%), while contiguity-preserving alternatives (pied-piping, fused-head construals, and the *big mess* family) occur at measurable rates (Reynolds, 2026d). Comprehension pushes the same way: a fronted *whose* or *which* defaults to a fused-head parse, so the discontinuous analysis also loses the parsing race (Culicover et al., 2022). The non-satiation of left-branch extraction, confirmed in a meta-analysis of 25 experiments (Lu et al., 2024), is exactly what a concentrated near-zero posterior predicts (§3.3.2).

Decaying-gain learning and logistic dynamics. The discrete update (45) is the primary dynamics. Applied to a fixed environment (exogenous evidence streams), it is a decaying-gain estimator: the posterior mean follows a hyperbolic trajectory of the form $a_0/(a_0 + b_0 + \bar{e}t)$, as plotted in Figure 4, and converges without bifurcating.

The logistic approximation $\dot{C} = r(g, c)C(1 - C)$ arises when the loop in §3.3.2 is closed: positive evidence is generated by production at a rate that grows with the current licensing state, while error and preemption mass accrue when the target is not chosen. Replacing those endogenous evidence streams by their expectations yields a logistic drift, with $r(g, c)$ a net evidence rate and a gain that is constant under bounded evidence memory. I use this ODE only for qualitative fixed-point and comparative-statics arguments; the fixed points at 0 and 1 belong to the closed-loop population dynamics, not to individual open-loop learning.

4.4 Emergent categoricality as bimodality

The sense of emergence used here is weak and mechanistic. Population-level categoricality is derived from learner-level updating, selection, and repair; it is not a label for whatever remains unexplained. This differs from Hopper's "emergent grammar", where grammar is perpetually provisional (Hopper, 1987). The present model allows conditioned states with near-absorbing attractors.

In the mean-field approximation, the closed-loop dynamics $\dot{C} = rC(1-C) + \kappa_{\text{rep}}\phi_{\text{rep}}(C^* - C)$ have attractors at the licensing extremes: when the net evidence rate r is positive, positive evidence dominates and the posterior is driven toward 1; when negative, error evidence and preemption mass drive it toward 0; and the repair flow, where present, steepens the basin around the community's convention C^*. Across construction nodes, a community whose evidence rates are bounded away from zero develops a bimodal distribution of licensing states: most nodes cluster near the attractors, and only low-evidence or contested nodes remain diffuse.

The mean-field picture is not a gesture. The occasion-level system–speakers sampling from the production rule, hearers updating discounted Beta posteriors–is a reinforcement process of Pólya type, and the relevant convergence technology is stochastic approximation for urn-type models (Pemantle, 2007). The signaling-game case, structurally the closest analogue, is treated directly: Argiento, Pemantle, Skyrms, and Volkov analyse micro-level reinforcement in the two-state case (Argiento et al., 2009), and Hu, Skyrms, and Tarrès extend the analysis (Y. Hu et al., 2011). Under those dynamics, an initial licensing asymmetry between competitors in a dense niche is amplified: the favoured variant's posterior concentrates near 1 and its rivals' near 0, with reversal probability decaying rapidly in the number of opportunities.

The claim as it applies here needs stating at its honest strength. What is proved is convergence for close analogues; what the paper needs is the same result under the present production rule, with the outside option at fixed utility and ascription-weighted evidence. I take that transfer to be theorem-adjacent–provable with standard stochastic-approximation techniques, at roughly $p \approx 0.8$–but it has not been carried out, and the bimodality claim inherits that status. Two consequences are worth accepting explicitly. First, selection intensity scales with $N_t(n,c) \cdot \Delta(d)$: update-configuring contrasts in dense niches converge fastest, which is the formal sense in which closed, high-opportunity, update-critical contrasts are exactly the parameter regime of emergent categoricality. Second, low-Δ contrasts face weak selection, so their licensing distributions are predicted to stay unimodal and gradient–policed like style, not like grammar–which is the §2 falsifier again, now derived rather than stipulated.

Sampling drift near the extremes makes the boundaries sticky in finite populations, while middle states stay vulnerable to renewed evidence. This connects the dynamics to complex-adaptive and cultural-evolutionary accounts of language change (Beckner et al., 2009; Blythe & Croft, 2012; Kirby et al., 2014), and to invisible-hand explanations, where macro-level order emerges as the unintended consequence of micro-level choices (Keller, 1994).

4.5 Derived obligatoriness

The state theory referred obligatory dimensions to $\text{OBL}(c, A)$ (§3.2) with a promissory note. Here it is paid: obligatoriness is not stipulated per language; it is the limit of preemption applied to zero-marking.

> A dimension d belongs to $\text{OBL}(c, \phi)$ for frame type ϕ iff, for every niche n in c with d at issue, every assembly leaving d unset has licensing driven to $C_t \approx 0$.

The derivation is the bimodality result applied with zero-marking as the disfavoured competitor. In a d-at-issue niche, the marked and unmarked assemblies compete; where the marked assembly wins every opportunity, the zero-marking assembly accumulates preemption mass exactly as any other rival does, and its posterior concentrates near zero. A dimension becomes obligatory exactly when that has happened in every niche where it is at issue. The claim is a corollary, and it inherits the bimodality result's status–conditional on that transfer, roughly $p \approx 0.85$.

The typological payoff is that §2.1's descriptions convert to predictions. English progressive marking is obligatory because simple-present assemblies in ongoing-at-issue niches have been driven to near-zero licensing; French zero-marking there has not been, so the dimension stays optional, and *Elle étudie maintenant* is fine. Turkish evidential marking is the same story on a different dimension. And grammaticalization's obligatorification gets a mechanism: a contrast on its way to obligatory status is one whose zero-marking competitor is losing dense niches one at a time, which predicts the intermediate stage–obligatory in some frame types and niches, optional in others–that grammaticalization corpora show.

4.6 Winnerless cells

Preempted gaps have a winning competitor. Defective paradigm cells do not, and the dynamics now say why the difference is stable rather than transitional.

Suppose candidates $f_1, \ldots, f_m$ for a dense cell are near-symmetric in initial licensing, the outside option (periphrasis, avoidance) has moderate fixed utility, and producing a weakly-licensed form carries a face cost. Then the production rule routes the cell's opportunities to the outside option: no candidate accumulates positive evidence. And here the attributability weighting (46) does its work: outside-option choices have $q_i \approx 0$, so no candidate accumulates much negative evidence either. Every candidate sits at low mean and low concentration–a self-maintaining interior state, metastable for long times, since escaping it requires some candidate to accumulate positive evidence that the avoidance attractor keeps starving it of.

Status, honestly graded: the qualitative mechanism–an avoidance attractor plus evidence starvation–is what the winnerless-cell example in §3.9 relies on, and I take it to be robust. The metastability claim as stated, persistence for times exponential in population size, is a conjecture with strong analogues in evolutionary dynamics, at perhaps $p \approx 0.6$ as stated and $p \approx 0.8$ that the qualitative mechanism survives formalization. Sims's (2023) point that the maintenance mixture varies across cases–structural uncertainty, lexical restriction, learning dynamics, avoidance–survives either way: the model supplies the avoidance-and-starvation component and is agnostic about the rest of the mixture.

The phenomenological payoff is now a derivation. Defective cells (**amn't*, *pobedit'* 1SG) show low mean with low concentration–"I don't know how to say it"–where preempted gaps show low mean with high concentration–"that's impossible". Same mean, different concentration, different feeling, read out by Φ (§3.7). The satiation-framing prediction follows: defective cells should behave like low-evidence forms, movable under framing; preempted gaps should not (Broadbent, 2009; Pertsova, 2016; Thoms et al., 2025).

4.7 Destabilization of moribund contrasts

Bounded memory buys one more result. With discount $\delta_m < 1$, the equilibrium concentration of a node's posterior is set by the balance of evidence inflow against forgetting: $\nu^* = O\big(N/(1-\delta_m)\big)$, with N the per-window opportunity count for the node's niches. If opportunity falls–a case distinction going moribund, a niche drying up–equilibrium concentration falls proportionally: licensing estimates drift toward their priors, and inter-speaker dispersion rises, because individual histories diverge when shared evidence thins. This is elementary given the discounted update, and I rate it accordingly ($p \approx 0.9$).

The corollary is a novel, testable leading indicator: for a contrast going moribund, judgment *dispersion* across speakers should rise measurably before mean judgments or production frequencies move. Dispersion leads; means lag. Variation corpora and judgment archives with repeated sampling over apparent time are exactly the data that could check it, and §5.2 states it as a registered prediction.

4.8 Actuation

The same dynamics define ACTUATION: a sign change in the expected evidence balance for licensing. Actuation occurs when the expected positive stream $s_t(u, c)$ begins to outweigh the combined negative streams $e_t(u, c) + p_t(u, c)$ across successive windows, so that the posterior mean $C_{t+1}(u, c)$ in (45) increases rather than decays. In the mean-field approximation from §4.3, the net evidence rate $r(u, c)$ becomes positive, yielding the familiar S-curve at the population level.

The factors that drive such bifurcations align with the motivations discussed in §2.5. Semantic reanalysis may increase utility (and thus $\rho_t^\star$) by making a form–value relation more transparent or useful. Social pressures may shift utilities (prestige effects) or alter the relevant community standard. Structural motivations such as analogical extension can systematically raise licensing probability by borrowing strength from frequent neighbors. Processing innovations may reduce error rates (e_t) or locality costs, improving the net evidence balance.

Individual innovation alone will not do. A few speakers adopting a marginal construction do not guarantee its success; the evidence balance has to shift so that acceptance systematically outweighs rejection across the relevant community. Many sensible innovations fail because they appear before those community conditions are in place.

Near the critical point, small perturbations in community attitudes can trigger rapid shifts between rejection and acceptance. As the balance becomes more strongly positive, the construction moves further from the unstable rejection state, making reversal increasingly unlikely. This produces the characteristic S-curve of documented language changes (Kroch, 1989). For constructions in the marginal zone, with near-zero evidence balance, the model predicts heightened sensitivity to external factors and greater cross-community variation. Small or peripheral communities may show volatile behaviour near bifurcation points before a change spreads to larger population centres.

The destabilization result sharpens the actuation picture at the other end of a form's life. Actuation is a sign change in the evidence balance; destabilization is a collapse in its concentration. A contrast can lose its grip not because negative evidence accumulates but because evidence of any kind stops arriving, and the model predicts the two routes are empirically distinguishable: actuation moves means along an S-curve with dispersion transiently high near the critical point, while moribundity raises dispersion first and moves means only as cohorts turn over.

4.9 Apparent exclusions without independent constraints

When a defined, saturated assembly exists but its pivotal node's licensing posterior is driven toward zero and concentrates there under persistent preemption in a dense opportunity set, speakers converge on a sharp, non-satiating rejection profile. Additional processing penalties in $R_i(u, c)$ (e.g. systematic garden-path repair due to an entrenched fused-head construal) can strengthen the subjective categoricality without any independent structural constraint being posited.

5 Theoretical implications

The payoff of OVMG isn't just a new decomposition of grammaticality. It says what the category should let us expect. If a form–value relation is grammatical in a communicative situation, it should pattern with other licensed resources in repair behaviour, transmission, satiation under exposure, and trajectories of change. If it is unlicensed, weakly licensed, or backed by little evidence, those expectations should fail in different, diagnostic ways (§3.3.2).

This is the sense in which the account is projectibility-first: grammaticality earns theoretical standing by the inferences it supports, not by the mere fact that a mechanism can be named for it (Boyd, 1999; Goodman, 1955; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). The profile that supports those inferences is conditioned form–value stability. The evidence offered here reaches two levels. First, bimodal licensing dynamics explain why some form–value relations stabilize near acceptance or rejection. Second, the production–repair loop explains how later use can help maintain that state. A stronger claim would require evidence for corrective control: a process that registers departures from the licensed profile and pushes the community back toward it. That stronger claim is left open (Reynolds, 2026b; Woodward, 2001, 2003).

This recasts the competence–performance relation. Licensing and feeling aren't two fundamentally different phenomena but different manifestations of the same conditioned state: processing and other performance factors help shape which form–value relations stabilize, and those stable relations in turn constrain performance. The identification strategy in §3.6 keeps this from becoming an escape hatch, since licensing and feeling are measured by different evidence streams and can falsify each other. The rest of this section develops the consequences: the account's reading of macro-typological universals, its distinctive predictions, and its relation to neighbouring frameworks.

5.1 Macro-typological constraints on grammatical design

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank's 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates.

A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of "narrow" word-order universals survive, whereas "broad" word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to "harmonic" combinations of features (for instance, consistent head–dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an OVMG perspective, these findings identify candidate attractors in the design space of operator-valued form–value relations rather than exceptionless grammatical laws. They don't by themselves show that a macro-typological feature is a language-internal grammatical object. The typological claim has to pass through a comparandum-to-realization mapping: one must specify which comparative feature is being tracked, how each language realizes the relevant operator value or form–value relation, and whether the resulting profile predicts held-out languages or diachronic trajectories

(Reynolds, 2025).

Under that discipline, supported patterns correspond to configurations for which the net evidence balance (positive evidence s_t vs. preemption/error mass $e_t + p_t$) in the dynamics of §4.3 is positive across a wide range of communities. They're cognitively and communicatively favourable enough that repeated episodes of change tend to push $C_t(u, c)$ toward fixation in lineage after lineage.

The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind complements OVMG by locating candidate stable regions; the present framework explains how local community dynamics and form–value coherence can make those regions reachable and persistent.

5.2 Predictions

The model's distinctive predictions come from separating positive evidence, opportunity structure, and counterfactual choice probability. Cross-linguistic gender remains a useful boundary case, but it isn't the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can be policed through paradigm inertia even when its public-update value is attenuated (Reynolds, 2026c). The sharper predictions concern preempted gaps, artificial learning, and L2 transfer.

Satiation framing, as a rank order. Framing manipulations are prior shifts, and a prior shift moves a posterior in proportion to $1/\nu$–the inverse of its concentration. The prediction is therefore not merely directional but ordinal across construction types: left-branch extraction $\approx$ **sheeps* (nil) $<$ entrenched-but-uncommon constructions $<$ independent relative *whose* $\approx$ **amn't* (largest). Participants told that tokens of a low-concentration form come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told the same tokens are errors; the same manipulation should leave left-branch extraction unmoved, as its non-satiation across 25 experiments already suggests (Lu et al., 2024). The dependent variables are rating change, confidence change (Φ), later repair behaviour, and elicited production.

Artificial-language learning. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target variant is never produced. In the high-opportunity condition, competitor variants repeatedly occupy the relevant niche and the target has high $\rho_t^\star$; in the low-opportunity condition, the niche rarely arises or the target would have low counterfactual utility. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. The design operationalizes the Figure 4 contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 preemption transfer. Early L2 rejections should track L1 preemption mass, not just L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, $q_h(c \mid e)$ and the constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's

(1972) concept of INTERLANGUAGE, but gives a more specific prediction: transfer is strongest where the L1 has accumulated large effective preemption mass.

DISPERSION AS A LEADING INDICATOR. For a contrast going moribund, the destabilization result (§4.7) predicts that inter-speaker judgment dispersion rises measurably *before* mean judgments or production frequencies move: falling opportunity thins the shared evidence, concentrations decay, and individual histories diverge while the average still holds. Dispersion leads; means lag. Variation corpora and judgment archives with repeated sampling over apparent time can test this directly, and the prediction is distinctive: accounts on which decline is driven by accumulating negative evidence predict means and dispersion moving together.

POLICING INTENSITY AS A PARAMETER. The repair stream (§4.3) converts the operator/style boundary from a classification into a regression: policing intensity should scale with the product of niche opportunity and update divergence, $N_t \cdot \Delta$, with both factors independently estimable–opportunity from opportunity-annotated corpora, divergence from repair-initiation rates in conversation-analytic corpora regressed on annotated update divergence. The §2 falsifier becomes quantitative: a closed, dense, non-configuring contrast has $\Delta \approx 0$ and should show style-like policing–a social-sanction baseline with gradient, unimodal licensing–however frequent it is. If repair rates track social indexing rather than Δ, the corrective term collapses into sociolinguistic policing and the controller claim fails.

CHANGE BY RE-LICENSING. Hard unification makes a commitment that a weighted-penalty model does not: no amount of entrenchment can offset an operator clash, so within-speaker gradience on compatibility failures like **I've finished it yesterday* should not exist. If perfect-plus-past-adverb combinations spread in a community, change has to proceed by *re-licensing*: a perfect construction with a different operator contribution enters the repertoire, predicting judgment distributions that are bimodal across communities and categorical within speakers, not gradual within-speaker cost-reduction. A soft-constraint model predicts the opposite signature. Judgment-distribution data from the relevant contact varieties decides it.

INTONATIONAL OPERATORS. Nothing in the model privileges segmental material, so intonational patterns are predicted to count as operator exponents wherever they form closed, high-opportunity, update-critical repertoires: the English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026c). The Turkish suffix-harmony analysis in Appendix A gives the parallel case for segmental exponent choice: phonological material matters for grammaticality when it realizes an operator exponent. The corresponding intonational test is whether mismatches between intonation and text (e.g. nuclear accent on a closed-class word under broad focus) are policed like grammar, with categorical rejection, repair, and resistance to framing, or like style.

5.3 RELATIONSHIP TO GENERATIVE GRAMMAR

Generative work made a central contribution by showing that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. Chomsky's *Colorless green ideas sleep furiously* made the point vivid: speakers can recognize syntactically well-formed sentences even when they're semantically bizarre. Generative analyses explain categorical constraints like the impossibility of extracting determiner-adjective sequences in English (**Which do you prefer car?*), and they capture the fact that such sequences remain ungrammatical even when their intended meaning is clear and processing demands are low.

OVMG keeps that lesson. Grammaticality judgments reflect systematic patterns (Dennett, 1991); those patterns can't be reduced to meaning or processing alone; and certain syntactic configurations appear to be categorically excluded regardless of context. The earlier diagnostics for center embeddings and preempted gaps build directly on ideas about recursive structure and acknowledge the generative discovery that some constructions appear to be excluded by grammar-internal constraints.

The same point governs the present account's use of formal evidence. A generative analysis can supply a real warrant when it specifies the target, diagnostics, evidence, and failure conditions for a grammatical contrast. OVMG isn't an attempt to replace every I-LANGUAGE analysis; it rejects only the stronger claim that categorical grammaticality must be explained by an autonomous syntactic component or a UG-backed inventory of primitives. Its target is the community-level licensing profile that makes some form–value relations categorically policed.

One well-known case is independent relative *whose* (1d). As Hankamer and Postal (1973) observe, the construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* can be used as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has driven theoretical development.

OVMG explains the contrast differently. Rather than positing an autonomous syntactic component, it treats grammatical constraints as products of form–value relations within specific communicative situations. For independent relative *whose* to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the natural way to package this information. The extreme rarity of contexts satisfying all these conditions appears to prevent the construction from becoming conventionalized as an English resource: speakers encounter it so rarely, despite perfectly common components, that even when all conditions align perfectly, the construction feels alien.

The difference is evidential rather than merely terminological. A generative theory has to explain why a syntactically possible and pragmatically useful construction is systematically avoided. OVMG treats extreme mismatches between predicted and observed frequency as possible evidence of strong negative weighting, even when the exact source of that evidence remains unclear. The result preserves the generative recognition of systematic constraints while embedding it in a theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are grammatical with one meaning but ungrammatical with another, such as (13) *I have 16 years*. While syntactically identical to grammatical expressions like *I have 16 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences.

OVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years has become conventionalized for expressing duration or future time (*I have 16 years until retirement*) but lacks positive evidence for expressing age, where a different construction (*I am 16 years old*) is the established pattern. Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that what gets down-weighted or up-weighted often depends on specific form–value associations rather than purely structural constraints.

5.4 Relationship to Construction Grammar

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) made a central contribution by treating learned form–meaning pairings as grammatical objects. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, include both individual morphemes and abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don't fit a clean core/periphery split.

CxG shows that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What's X doing Y?* construction carries an implication of incongruity that can't be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

OVMG shares these CxG commitments about form–value relations and constructional abstraction. It doesn't privilege one expressive substance over another. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026c). The OVMG claim is narrower. Categorical grammaticality clusters around form–value relations with a particular role: they're high-opportunity, closed-paradigm contrasts that configure update, role allocation, scope, reference, or repair.

The contrast becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, clause-linking marker, or obligatory allomorph can mis-set the update instruction even when the intended message is recoverable. The age-predication example (*I have 25 years*) is transparent but not licensed for that English frame; its rejection comes from non-licensing in a high-opportunity constructional niche, not from the substance of syntax or morphology as such.

OVMG preserves CxG's systematic treatment of constructions while adding a prediction about which constructional contrasts attract categorical policing. The relevant contrasts are those whose closure and opportunity structure drive the licensing posterior toward the bimodal regimes in §4.4. That lets CxG's continuum of constructions coexist with a non-arbitrary explanation of why some violations are heard as grammar and others as style, stance, or appropriateness.

5.5 Relationship to usage-based approaches

OVMG shares with usage-based approaches (UBA) (Bybee, 2006, 2007, 2010) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Several pieces of the formal core come directly from this tradition. The individual/population split ($\Lambda_{i,t}$ for a speaker's own licensing state, C_t for the population rate) restates Schmid's (2020) separation of ENTRENCHMENT (the cognitive reorganization of a speaker's knowledge) from CONVENTIONALIZATION (the social establishment of a community's regularities), the two coupled feedback cycles of his Entrenchment-and-Conventionalization model. The population dynamics themselves descend from Croft's (2000) utterance-selection theory, in which variants are replicated and selected within a community; the S-curve results the model reuses are already worked out in that lineage (Blythe & Croft, 2012).

Frequency per opportunity is likewise not new: it's the core of Stefanowitsch's (2006) case that corpora contain negative evidence, where a form counts as "significantly absent" rather than "accidentally absent" only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov's principle of accountability (Labov, 1972).

OVMG adds their joint formalization and a prediction. The licensing-versus-selection factorization $\pi_t \approx C_t \cdot \rho_t^\star$ turns Croft's replication-versus-selection into an estimable product; effective preemption mass $p_t = N_t \cdot \rho_t^\star$ puts a quantity on the absence that Stefanowitsch left open, since he showed how to establish that a form is significantly absent but stressed that significant absence "does not, in itself, provide any clues as to why". Licensing-versus-selection is exactly that missing clue: a significantly absent form may be unlicensed ($C_t \approx 0$) or licensed but dispreferred ($\rho_t^\star \approx 0$), and the two project differently. The Beta dynamics then convert these quantities into the bimodality result (§4.4) and the prediction about which contrasts attract categorical policing. The relation to Goldberg's (2019) COVERAGE and statistical preemption is one of formalization in the same spirit: her "explain me this" puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a preemption-mass story in which the competitor wins a high-opportunity niche. One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single construction-competition process. OVMG keeps them analytically separate: error evidence e_t and preemption mass p_t are distinct streams, but they enter the Beta update additively (§4.3), so the model can represent Ambridge and colleagues' single-process view as their sum while leaving the separability an empirical question rather than a stipulation.

The analysis of independent relative *whose*, as in ?*I saw Joan, a friend of whose was visiting*, is a case in point. A simple UBA account might predict that this construction's marginality stems from its low frequency. But this explanation proves insufficient: the construction is rare, and it's dramatically rarer than one would expect given the frequency of its component parts. Independent *whose* appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative more common than observed.

Raw component frequency is therefore the wrong predictor; *whose* isn't a preempted gap. The frequency of independent and relative *whose* taken separately doesn't create opportunities for the niche that requires both at once, together with an accessible possessor, a predictable possessum, and a licensing ellipsis site; that convergence is vanishingly rare. This isn't the significant absence of a preempted form, which needs a large opportunity set and a competitor that keeps winning; it's the low-opportunity uncertainty described in the diagnostic categories above. The tension between "rarer than expected" and "uncertain rather than rejected" dissolves once licensing is separated from selection: *whose* is under-*observed* because its niche rarely arises, not preempted because a rival repeatedly beats it.

OVMG doesn't exempt the operator stratum from usage dynamics; it makes those dynamics more specific. A rare form can remain grammatical when the opportunity set is tiny or when higher-grain constructions license the token by partial pooling. A transparent candidate can become categorically rejected when the opportunity set is large, the analogical candidate would have had high $\rho_t^\star$, and an entrenched competitor wins every time. The result preserves usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption.

5.6 Relationship to logicality of language accounts

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del Pinal, 2019; Fox, 2000). This "logicality of language" hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are "L-trivial", meaning their triviality arises solely from the meaning and configuration of logical or closed-class terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009). A key challenge is explaining why simple tautologies or contradictions (e.g. *It's raining and it isn't raining*) are often acceptable. Del Pinal (2019) argues against "Logical Skeletons" (which assume the system ignores open-class word identity) in favour of "LF+RESCALE", a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second "raining" as "raining hard") to yield non-trivial meanings.

OVMG offers a broader account of ungrammaticality. "Logicality" approaches explain restrictions tied to logical and closed-class vocabulary through L-triviality. OVMG also covers cases less easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), and extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions while separating hard compatibility from posterior licensing, OVMG accommodates cross-linguistic variation and degrees of acceptability, which are less central to the L-triviality filter.

LF+RESCALE provides a specific mechanism for acceptable "trivialities". OVMG treats those cases as possible products of more general interpretive principles within community norms, without positing a dedicated deductive module or specific operators like RESCALE. It also keeps objective grammatical status separate from subjective processing effects or "feelings" (§3.7). OVMG frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.4, rather than a direct output of logical computation within syntax.

5.7 Relationship to relevance-theoretic accounts

Recent work by Scott-Phillips (2026) makes a useful observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from

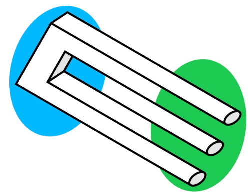

Figure 5: Impossible trident

an inherent impossibility of interpreting an utterance consistently with these presumptions, much as an impossible trident (Figure 5) can't be interpreted as physically cohesive in any context.

OVMG shares several premises with this account. Both reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems, and both treat language as developing from communicative needs rather than autonomous syntactic principles. OVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, OVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. OVMG instead analyzes how the present perfect's tense–aspect operator value clashes with the meaning of the lexeme *yesterday*. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

Empirically, OVMG requires tests of whether specific form–value relations are stable within a communicative situation and where operator-value conflicts arise. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. The clearest comparison would test the two accounts on novel constructions as they become acceptable or unacceptable.

6 Limitations and future directions

The account has real limits, and each points to work it invites rather than forecloses. One is its reliance on the concept of "communicative situation". Speakers move among overlapping contexts that shift with the immediate setting, durable social positions, and the norms they orient to. The account builds in this fluidity, but operationalizing these dimensions, and measuring their influence on grammatical stability, will take more precise methods. A fourth limit is mathematical candour: the bimodality result that categoricality rests on is proved for close analogues of the present dynamics, not for the present dynamics themselves (§4.4), and the framework's central emergence claim inherits that theorem-adjacent status until the transfer is carried out.

A second boundary concerns stylistic variation and individual preference. Most choices that fall within grammatical acceptability are not cases of (un)grammaticality at all. They matter for this account only when they become tied to communicative situations strongly enough to affect licensing, repair, or categorical rejection. The account still has to explain how stylistic preferences become grammatical licensing facts.

A third concerns purely formal constraints that seem to exist independently of meaning or communicative role. The paper argues that many apparently arbitrary restrictions follow from historical processes and competing motivations, but there may be residual cases that resist even that treatment, and they'd sharpen the boundary of the account.

6.1 Methodological considerations

The development of experimental syntax since the 2000s (Schütze, 2016) has transformed how gradient acceptability is measured, letting researchers quantify subtle differences in judgments and track how they change with exposure. The framework's predictions invite corpus-based, experimental, and cross-linguistic investigation. Several avenues stand out:

1. *Corpus analysis:* Large, balanced corpora allow researchers to track frequency patterns and stability over time. Investigating rare or marginal forms (e.g. the independent relative *whose*) in corpora can reveal whether low frequency corresponds to genuinely unstable operator-value patterns or merely a sampling artifact. Comparative corpus research in multiple languages can test predictions about how community values influence grammatical distinctions.
2. *Grammaticality judgment experiments:* Controlled psycholinguistic tasks, including magnitude estimation or forced-choice paradigms, can assess subtle gradience in grammaticality judgments. By manipulating exposure and increasing familiarity with marginal forms, researchers can measure satiation effects and distinguish between entrenched categorical constraints and constructions that become more acceptable through repeated exposure.
3. *Processing measures:* Eye-tracking, self-paced reading, and EEG studies can identify whether some judgments arise from processing overload rather than durable grammatical constraints. If a form becomes easier to parse with practice, it suggests that processing complexity, rather than an immutable rule, lies behind initial judgments of ungrammaticality.
4. *Sociolinguistic fieldwork:* Investigations in communities with distinct dialects or multilingual practices can clarify how social and pragmatic motivations shape grammatical stability. Elicitation tasks and careful comparisons across speech communities can confirm that what counts as grammatical depends on local norms rather than universal principles.
5. *Longitudinal and historical studies:* Diachronic corpora and historical grammars can trace how marginal constructions evolve, testing predictions about which factors lead certain patterns to stabilize, which fade, and which undergo reanalysis. Tracking a community's acceptance of previously dubious constructions over time provides direct evidence for the gradual entrenchment processes posited by the framework.
6. *A threshold-discontinuity test:* The categorical read-out earns standing only if community treatment of a form changes *discontinuously* as the posterior mean crosses $\tau(c)$, over and above the smooth effect of $\mathbb{E}[G_t]$. That's a regression-discontinuity question. Hearer responses are naturally multinomial (accept, repair, request clarification, sanction), and covariate-adjusted local-likelihood estimators for discontinuities in multinomial response probabilities exist off the shelf (Lazzaro, 2026), with the processing-cost components of R_i entering as predetermined covariates. Two caveats are design constraints rather than afterthoughts: in a confirmatory test, $\tau(c)$ is latent,

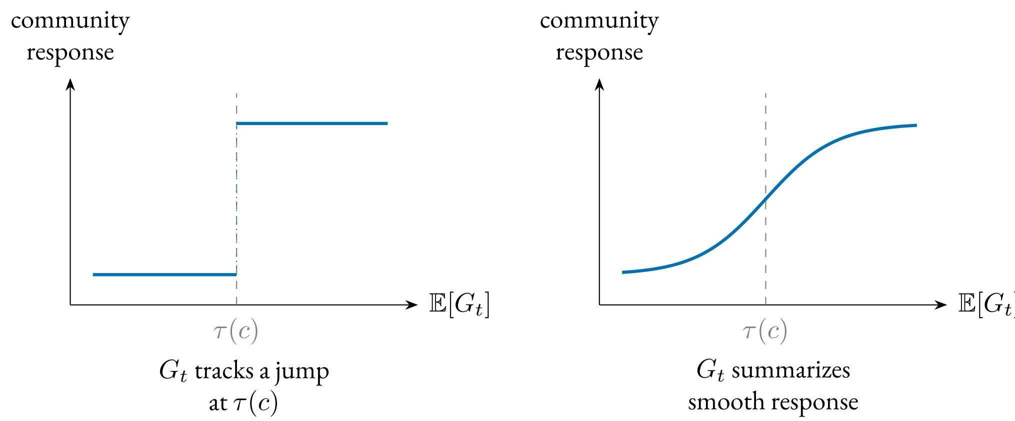

Figure 6: The threshold-discontinuity test. If categorical grammatical status tracks a discontinuity in the population state, community treatment of a form should change discontinuously as $\mathbb{E}[G_t]$ crosses the situational threshold $\tau(c)$ (left); if the read-out only summarizes the posterior mean, treatment should vary smoothly through it (right). A covariate-adjusted regression-discontinuity design on multinomial hearer responses (Lazzaro, 2026) distinguishes the two.

so it has to be estimated and registered before the outcome data is touched; and the bimodality result (§4.4) predicts that the density of forms near the threshold is thin, so power is scarce exactly where the test bites. The design also yields a further distinctive prediction: because $\tau(c)$ is decision-theoretic (§3.3), independently manipulating the stakes should move the location of the discontinuity. Smoothness through the threshold, by contrast, would leave the categorical read-out as bookkeeping over the posterior mean.

Figure 6 states the test in one picture: the categorical predicate tracks a discontinuity only if community treatment jumps at $\tau(c)$.

7 Conclusion

Grammaticality, de-idealized, isn't a categorical fact about an ideal speaker in a homogeneous community but a conditioned form–value relation. Its profile–whether some assembly covers the form, whether the assembly's operator contributions unify, whether its obligatory dimensions are saturated, and whether the population licenses its constructions–is specified relative to a communicative situation and read through a posterior that carries concentration as well as mean. The old competence–performance split becomes two layers with separate evidence streams, a conventional, population-level status and a two-channel subjective read-out–anomaly and confidence–identified by different measures so that they can falsify each other (§3.6). Categorical grammaticality is then derived rather than stipulated, an emergent bimodality of the licensing state under preemption (§4.4), which leaves independent structural impossibility as a residual case rather than the default explanation.

What the profile is for is projection. A category earns its standing by supporting expectations licensed by partial observation, not by the mere presence of mechanisms: repair, transmission fidelity, satiation, and trajectories of change, over the licensing state as such, its concentration as well as its mean (§3.3.2). The supporting claims are graded separately. Bimodal dynamics stabilize the profile, and the production–repair loop maintains it. The repair stream supplies a candidate corrective system–a process that registers operator mis-settings and pushes the licensing state back–and states the intervention signature that would earn it controller status, along with the finding that would defeat it (§4.3). The stronger claim is thereby made testable rather than merely left open. On this account, grammaticality matters because it predicts which forms will be repaired, transmitted, softened by exposure, or resistant to change.

These commitments expose the framework to failure. The account predicts that satiation should be more readily inducible for low-evidence constructions than for high-opportunity preempted gaps, that artificial-language learners should distinguish zero evidence from strong preemption, and that early second-language rejections should track first-language preemption mass rather than second-language frequency alone. Most sharply, community treatment of a form should change discontinuously as licensing crosses the situational threshold; otherwise, the categorical predicate adds no explanatory content beyond the graded score (§6.1).

The account doesn't have the simplicity of a one-variable theory. It doesn't reduce grammaticality to a single scalar, a competence–performance split, or a social-pragmatic explanation. Its economy is architectural: the same recurring distinctions (coverage, compatibility, saturation, licensing, subjective feeling, opportunity structure, and diachronic support) handle cases that otherwise invite separate stories. That economy has to be earned empirically. The conditions need independent indicators, and the projections need to fail when those indicators are wrong.

It also carries a significant memory burden. Speakers need not store every utterance as a separate item, but the account asks them to retain many situation-indexed pairings among forms, operator values, constructional neighbours, opportunity estimates, competitors, and repair histories. That inventory is large, but the revision converts it from a liability acknowledged into the model's data structure: the situation-indexed pairings among forms, values, neighbours, opportunity estimates, competitors, and repair histories are exactly the sufficient statistics the licensing posterior runs on (§§3–4), and the hierarchical pooling across construction families, lexemes, and situations is what makes the burden estimable rather than merely assumed. If linguistic memory is presumed small in advance, stored generalizations get forced into a rule-plus-exception format, and conditioned licensing is misdescribed as residue.

The program these predictions define is measurement-first. Estimate licensing from converging non-rating indicators (corpus rate per opportunity, elicited production, repair), fix the conditioning state before predicting rather than after, and prioritize cross-linguistic contrasts whose operator status differs, since that's where the account predicts categoricality should form or dissolve. De-idealizing grammaticality trades a tractable idealization for a profile that answers to evidence. It buys a category that predicts: which rejections soften, which forms transmit, and which absences hold.

A Turkish vowel harmony and operator-exponent realization

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.[3] Only suffixal harmony matters for OVMG, and even there the role is indirect. Vowel backness is not a public-update value; plural and past are. In Turkish, the exponents of those operators must satisfy suffixal harmony.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* 'doctor' violate backness harmony, but are fully acceptable; speakers store the form, and the community-level licensing state remains near 1.

(49) doktor
doctor
'doctor' (disharmonic stem, grammatical)

Because the word contains no malformed operator exponent, a stored construction covers it, the relevant operators unify, and $G_t \approx 1$.

SUFFIXAL HARMONY: OPERATOR-EXPONENT REQUIREMENT. Inflectional morphemes are lexicalized with an underspecified vowel; the correct allomorph has to copy [±BACK] (and, for some suffixes, [±ROUND]) from the final stem vowel. Using the "wrong" vowel leaves the exponent-selection requirement unsatisfied, so no well-typed suffixal assembly covers the intended form.

(50) a. kitap-lar
book-PL.BACK
'books' (harmonic, grammatical)

b. * kitap-ler
book-PL.FRONT
intended 'books' (suffix harmony violation)

(51) a. gül-dü
laugh-PST.FRONT.ROUND
's/he laughed' (harmonic, grammatical)

b. * gül-du
laugh-PST.BACK.ROUND
intended 's/he laughed' (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- The plural and past constructions are licensed at $C_t(\kappa, c) \approx 1$ for every speaker: what is entrenched is the node, not the ill-formed token.
- Their form templates lexicalize an underspecified vowel with a harmony condition: the exponent copies [±back] (and, for some suffixes, [±round]) from the final stem vowel. A mismatched allomorph satisfies no template, so no well-typed assembly covers the string: $\mathcal{A}(f, v) = \emptyset$ and $G_t = 0$ exactly–a structural failure at exponent selection, with $\Phi \approx 1$.

[3] See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arık, 2015) for experimental evidence on native judgments.

So speakers judge the word categorically wrong, not just odd-sounding, and the confidence is structural rather than evidential. This puts suffix-harmony errors in the same broad status class as word salad–coverage failure–but through a different route: a typed exponent-selection template rejects the former, while the latter lacks a coherent covering assembly altogether. By contrast, *doktor* in (49) is covered by its own stored node: stem disharmony violates no exponent-selection condition, so $G_t \approx 1$ and the disharmony registers only as a phonological cost in the feeling channel R_i (§3.7).

Localized exceptions. Some derivational suffixes (e.g. *-imsi* '-ish') are lexically marked "disharmonic". Speakers store them, and community-level licensing remains near 1: no exponent condition is violated, so $G_t \approx 1$ despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

Turkish suffix-harmony errors are therefore grammaticality failures in a precise sense: they do not mis-set a public-update contrast themselves, but they leave a closed inflectional operator with no well-formed exponent, which is a failure of coverage, not of licensing. Stem disharmony lowers phonological well-formedness and affects only the feeling channel. A counterexample would be a case where phonology alone produced categorical, framing-resistant rejection without any operator-exponent requirement in play.

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